GLOBAL STABLE SOLUTIONS OF THE ONE-PHASE BERNOULLI PROBLEM IN DIMENSION SIX

ABSTRACT. We prove that every global classical stable solution of the one-phase Bernoulli problem in $\mathbb{R}^6$ is one-dimensional. We use OpenAI's algebraic certificate, checked by exact rational computation, to exclude nonflat smooth stable cones in dimensions five and six. This lets us apply the blow-down theorem of Fernández-Real and Serra, which gives mean-square gradient closeness to two-sided linear profiles on large balls for a hypothetical nonflat solution. Our local gradient-defect estimate converts the averaged error into a supremum bound for the defect. The direction-independent length defect then vanishes at each fixed point, giving $|Du| = 1$ in the positive phase. Harmonicity then gives $D^2u = 0$. We also obtain local curvature bounds and axial symmetry for finite-index solutions.

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1. Introduction

Let $u \geq 0$ be continuous in an open set $A \subset \mathbb{R}^n$. Write $\Omega = \{u > 0\}$ and $\Gamma = \partial\Omega \cap A$. The one-phase Bernoulli problem is

$$\Delta u = 0 \quad \text{in } \Omega, \qquad u = 0, \quad |Du| = 1 \quad \text{on } \Gamma. \tag{1.1}$$

For a classical solution, Γ is smooth, Ω lies locally on one side, and u is smooth up to Γ from Ω. Since $u = 0$ on Γ, its tangential derivatives vanish. Its inward derivative is nonnegative by positivity and equals one by (1.1). Thus the positive-side trace of Du is the inward unit normal N. We write $\nu = -N$ for the outward normal and $H = \operatorname{div}_\Gamma N$ for the sum of the principal curvatures with the inward convention.

For $\phi \in C_c^\infty(A)$, integration by parts on the positive phase gives

$$\int_A u\,\partial_i\phi = \int_\Omega u\,\partial_i\phi = -\int_\Omega (\partial_i u)\phi,$$

since the boundary term contains the zero trace of u. The ambient weak gradient is therefore Du in Ω and zero elsewhere. Smoothness up to Γ makes it locally bounded, so u is locally Lipschitz. A global solution has $A = \mathbb{R}^n$.

On an open set D compactly contained in A, the Alt–Caffarelli energy is $J(u; D) = \int_D (|Du|^2 + \mathbf{1}_\Omega)$ [1]. Take a smooth family of diffeomorphisms Φ_t with $\Phi_0 = \mathrm{id}$, equal to the identity outside a fixed compact subset of D. Stability means that the second derivative at zero of $J(u \circ \Phi_t^{-1}; D)$ is nonnegative for every such D and Φ_t. For classical solutions we use the equivalent scalar inequality

$$Q(\phi) := \int_\Omega |D\phi|^2 - \int_\Gamma H\phi^2 \geq 0 \tag{1.2}$$

for $\phi \in C_c^\infty(A)$. See [4, Appendix A, (A.1)] for the nonnegative-test formulation on a ball. Appendix A proves the signed inequality and the converse. Boundary integrals use surface measure and tensor norms are Euclidean.

A logarithmic cutoff argument gives one-dimensionality for entire classical stable solutions in the plane [12, Theorem 3.1]. Chan, Fernández-Real, Figalli, and Serra proved rigidity in dimension three [4, Theorem 1.5], and Fernández-Real and Serra proved it in dimension four [9, Theorem 1.1].

Theorem 1.1 (Global classical stable rigidity)**.** *Let u be a global classical stable solution of the one-phase Bernoulli problem in $\mathbb{R}^6$. Then $D^2u = 0$ in Ω. If Γ is empty, u is a nonnegative constant. Otherwise there is a unit vector e such that u has one of the following forms, where $t_+ = \max(t, 0)$:*

$$u(x) = (e \cdot x - a)_+ \qquad \text{for some } a \in \mathbb{R},$$
$$u(x) = (e \cdot x - b)_+ + (a - e \cdot x)_+ \qquad \text{for some } a < b.$$

In Figure 1, the gradients on the positive pieces are e for the half-space and $-e, e$ for the two sides of the strip. These affine pieces are harmonic and have unit gradient. The boundary planes have $H = 0$, so $Q(\phi) = \int_\Omega |D\phi|^2 \geq 0$ and both profiles are stable by Appendix A.

1.1. Research context.

Weiss's monotonicity formula relates singularities of minimizing free boundaries to nonflat one-homogeneous minimizing cones [14]. On a ball $B \Subset A$ in dimension three, the functional of Caffarelli, Jerison, and Kenig is $J(\cdot\,; B)$ when the smooth phase costs are $Q_+ \equiv 1$ and $Q_- \equiv 0$, so the required positive gap is $Q_+ - Q_- = 1$. If $u \geq 0$ minimizes J among nonnegative competitors with its trace on ∂B, it also minimizes among signed competitors with that trace. Indeed, replacing a competitor v by $v^+ = \max\{v, 0\}$ preserves the trace and the positive set and can only decrease the Dirichlet energy. Their Corollary then gives smoothness of the interior free boundary via half-space blow-ups [3, Corollary, p. 84, and proof, pp. 92–93].

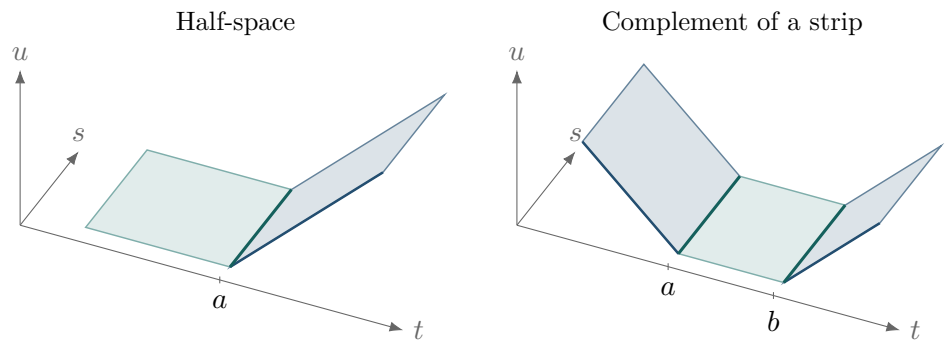


FIGURE 1. Graphs of the two nonconstant profiles over a slice with $t = e \cdot x$ and transverse coordinate s. The height u is constant along s. The pale teal patches mark the zero phase, and the dark teal edges mark the free-boundary traces.

For a local J-minimizer in dimension four, a blow-up at an interior free-boundary point is a nonzero one-homogeneous global minimizer with the origin on its free boundary [13, Proposition 2.1]. A blow-up of this cone at a nonzero free-boundary point splits off the radial line, leaving a nonzero one-homogeneous minimizer in three dimensions [14] [13, Lemmas 2.2 and 2.3]. The factor vanishes at the origin and is positive along a ray approaching it. The three-dimensional regularity result therefore makes it classical at that free-boundary point. Its blow-up there is the unit half-space with the same inward normal, by the one-sided first-order expansion at the smooth boundary. Since the factor is one-homogeneous, it equals that blow-up. The off-vertex blow-up of the four-dimensional cone is therefore a half-space, and flatness regularity [1] makes the cone classical at the chosen point. This holds at every nonzero free-boundary point of the cone. Minimality gives stability under compactly supported deformations away from the vertex, so Jerison and Savin's Theorem 1.1 makes the cone a half-space. Flatness regularity at the original point gives smoothness of the free boundary, the dimension-four conclusion of their Corollary 1.2 for the same phase costs $Q_+ \equiv 1$ and $Q_- \equiv 0$ [11, Theorem 1.1 and Corollary 1.2].

For their one-homogeneous cones $U_{n,k}$ with $O(n-k) \times O(k)$ symmetry and $1 \le k \le n-2$, Firester, Tsiamis, and Wang record instability for $3 \le n \le 6$, citing computations by Caffarelli–Jerison–Kenig and Hong, and prove strict stability for $n \ge 7$. Here strict stability means that the first Robin eigenvalue on the spherical link exceeds the Hardy threshold $-(n-2)^2/4$ [10, Theorem 1.1 and Section 2.1].

Every nonzero one-homogeneous global minimizer of J in dimensions $1 \le n \le 6$ is a half-space solution [13, Theorem 1.1]. The existence part of that theorem uses De Silva and Jerison's nonflat one-homogeneous global minimizer U_0 of the same energy in $\mathbb{R}^7$ [5, Section 1 and Theorem 1.1]. Thus seven is the first dimension admitting a nonflat minimizing cone. The positive phase of U_0 is

$$\mathcal{C} = \{(x', x_7) : |x_7| < c|x'|\}, \qquad c > 0.$$

Away from the origin, the free boundary consists of the analytic graphs $x_7 = \pm c|x'|$ over $x' \neq 0$. A tangent hyperplane at the origin would have to contain every boundary ray, but the ray directions $(e_i, \pm c)$, for the coordinate vectors e_i of $\mathbb{R}^6$, span $\mathbb{R}^7$. Hence the origin is singular. The cone $\mathcal{C}$ is connected because every point can be joined vertically to the connected punctured hyperplane $\{x_7 = 0,\ x' \neq 0\}$.

For sharpness of Theorem 1.1 in the classical class, apply De Silva, Jerison, and Shahgholian's theorem to the minimizing cone U_0 with its isolated singular vertex. They give an upper global J-minimizer $U \ge U_0$ with analytic free boundary at distance one from the origin [6, Theorem 1.1(i)–(ii)]. Their proof gives a local Lipschitz graph with the positive phase on one side. One-phase

boundary regularity then gives a classical solution of (1.1) [6, proof of Theorem 3.6]. The full family of rescalings $R^{-1}U(R\,\cdot)$ converges locally uniformly to U_0 as $R \to \infty$ [6, Lemma 3.1]. Minimality makes U stable, since each compactly supported deformation is an admissible competitor in a ball containing its support. If $D^2U = 0$ in $\{U > 0\}$, then U is affine on the connected subset $\mathcal{C} \subset \{U > 0\}$, say $U(x) = a \cdot x + b$ there. For $p = (p', 0)$ with $p' \neq 0$, both Rp and $-Rp$ lie in $\mathcal{C}$ for $R > 0$. Thus the blow-down gives

$$U_0(p) + U_0(-p) = \lim_{R\to\infty} \frac{U(Rp) + U(-Rp)}{R} = \lim_{R\to\infty} \frac{2b}{R} = 0.$$

Since $U_0(\pm p) > 0$, this is a contradiction. Thus U has nonzero Hessian somewhere in its positive phase. See [7] for a survey of the minimizing and stable theories.

1.2. **The gradient defect.** Writing $t = e \cdot x$, the strip profile in Theorem 1.1 rescales to

$$\frac{u(Rx)}{R} = (t - b/R)_+ + (a/R - t)_+.$$

The zero strip has width $(b-a)/R$. Since the positive-part map is 1-Lipschitz, the rescaled profile differs from $t_+ + (-t)_+ = |t|$ by at most $(|a| + |b|)/R$. The limiting slopes e and $-e$ motivate a defect that vanishes at both. With $P_e p = p - (p \cdot e)e$, set

$$h_e = \left(|P_e Du|^2 + \frac{(1 - |Du|^2)^2}{30} \right)^{1/2} \quad \text{in } \Omega.$$

The first term under the square root vanishes when Du is a multiple of e, and the second vanishes when $|Du| = 1$. Thus $h_e = 0$ precisely at $Du = \pm e$. The next estimate converts small mean-square defect on a ball into pointwise control on the ball of half the radius.

Theorem 1.2 (Local gradient-defect estimate)**.** *There are dimensional constants $C, \varepsilon_0 > 0$ with the following property. Let u be a classical stable solution in $B_R(x_0) \subset \mathbb{R}^6$, with $|Du| \le 1$ in its positive phase in this ball. For a unit vector e put*

$$E_e = \left(R^{-6} \int_{\Omega \cap B_R(x_0)} h_e^2 \right)^{1/2}.$$

If $E_e \le \varepsilon_0$, then

$$\sup_{\Omega \cap B_{R/2}(x_0)} h_e \le C E_e.$$

1.3. **Applications of rigidity.** To apply Theorem 1.1 in lower dimensions, extend the solution constantly in the extra coordinates. This adds only zero principal curvatures to the free boundary. Slicing an ambient test in those coordinates recovers the original stability form and adds a nonnegative Dirichlet term. The product argument is given in Section 5.

Corollary 1.3 (Rigidity in dimensions at most five)**.** *Let $1 \le n \le 5$ and let u be a global classical stable Bernoulli solution in $\mathbb{R}^n$. Then $D^2u = 0$ in Ω. If Γ is empty, u is a nonnegative constant. If $\Gamma \neq \varnothing$, u has one of the two forms in Theorem 1.1, with e a unit vector in $\mathbb{R}^n$.*

Local Hessian and curvature bounds were proved in dimension three [4, Corollary 1.7] and dimension four [9, Corollary 1.3]. In Section 6 we rescale at points of large Hessian. The global limit retains norm one either for its interior Hessian or for the Hessian trace from one positive side of a limiting sheet. If the limit is classical, rigidity makes it affine on each positive component. If two limiting sheets touch, the limit is an absolute-value profile, affine on each positive side. Both alternatives contradict the norm-one limit. The curvature bound follows from the Hessian bound because $D^2u(X, Y) = \langle D_X N, Y \rangle$ for tangent vectors X, Y on Γ.

Corollary 1.4 (Local Hessian and free-boundary curvature estimates)**.** *Let $1 \le n \le 6$ and let u be a classical stable solution in $B_{2R}(x_0) \subset \mathbb{R}^n$, with $x_0 \in \Gamma$. Then*

$$\sup_{\Omega \cap B_R(x_0)} |D^2 u| \le \frac{C_n}{R}, \qquad \sup_{\Gamma \cap B_R(x_0)} |\mathrm{II}_\Gamma| \le \frac{C_n}{R}. \tag{1.3}$$

Here C_n depends only on the dimension and II_Γ is the second fundamental form. For $n = 1$, $u'' = 0$ on the positive intervals and the second inequality is vacuous.

The scalar Morse index is the supremum of the dimensions of subspaces of $C_c^\infty(\mathbb{R}^n)$ on which Q is negative definite.

Finite index implies stability outside some ball. Otherwise we could choose k tests with $Q(\phi_i) < 0$, each supported outside a ball containing all the earlier supports. The supports are disjoint in both terms of (1.2), so

$$Q\left(\sum_{i=1}^k a_i \phi_i\right) = \sum_{i=1}^k a_i^2 Q(\phi_i) < 0 \qquad \text{for } (a_1, \dots, a_k) \neq 0.$$

This produces negative subspaces of arbitrarily large dimension.

Basulto and Kamburov classified nonconstant entire classical solutions in the plane with finite Morse index and connected positive phase [2, Theorem 1.1]. Fernández-Real, Florit-Simon, and Serra proved axial symmetry in dimension three and obtained a higher-dimensional classification under rigidity hypotheses for entire stable solutions and smooth stable cones [8, Theorems 1.1 and 1.2]. Section 6.2 shows that the two corresponding critical dimensions are at least seven, so their classification applies to finite-index solutions through dimension six.

Corollary 1.5 (Finite-index axial symmetry)**.** *Let $3 \le n \le 6$. Every entire classical Bernoulli solution of finite scalar Morse index is axially symmetric about a line. If u cannot be written as a function of one linear coordinate, then, after a translation, rotation, and Bernoulli scaling, it has one of the following forms.*

1. *The exterior-ball solution. It is zero on $\{|x| \le 1\}$ and equals $(1 - |x|^{2-n})/(n-2)$ on $\{|x| > 1\}$.*
2. *A solution $u(x', x_n) = U(r, x_n)$, $r = |x'|$, that is even in x_n, with $u_{x_n} > 0$ in $\Omega \cap \{x_n > 0\}$ and $U_r < 0$ in $\Omega \cap \{r > 0\}$. Its zero phase is*

$$\{(x', x_n) : r \ge 1,\ |x_n| \le g(r)\}.$$

The function g is continuous on $[1, \infty)$, smooth on $(1, \infty)$, and satisfies $g(1) = 0$, $g'(r) > 0$ for $r > 1$, and $\lim_{r \to \infty} g(r) \in (0, \infty)$.

1.4. **Proof overview.** Let n_* be the first dimension $n \ge 2$ admitting a global one-homogeneous stable solution, classical away from the origin, with a free boundary through the origin and distinct from the half-space profiles [9, Definition 4.1]. Section 2 uses the stability inequality on the spherical link to exclude such cones in dimensions five and six. Combined with the lower-dimensional classification, this gives $6 < n_*$.

For a solution with nonzero Hessian, center a free-boundary point at the origin and rescale to arrange $\|D^2 u\|_{L^\infty(\Omega \cap B_1)} \ge 1$ as in Section 5. With $6 < n_*$, the gradient term of the blow-down estimate [9, Proposition 4.12, (4.12)] gives directions e_R and profiles $V_R(x) = |e_R \cdot x|$ such that

$$R^{-6} \int_{B_R} |Du - DV_R|^2 \longrightarrow 0 \qquad \text{as } R \to \infty.$$

The absolute-value profile has a positive phase on both sides of its ridge, so it falls outside the classical cone exclusion [7, Remark 4]. Every global classical Bernoulli solution in dimension $n \ge 2$ satisfies $|Du| \le 1$ in its positive phase [4, Lemma 3.2]. With this bound, the squared gradient error from either unit slope of V_R controls $h_{e_R}^2$ on the positive phase by (5.2). Thus $E_{e_R} \to 0$.

For the local estimate, Section 3 shows that the regularized defect $h_\delta = (h_e^2 + \delta^2)^{1/2}$ satisfies $h_\delta \Delta h_\delta \geq |D^2 u|^2/75$. In Section 4, dividing h_δ by $b = (1 + |Du|^2)/2$ cancels the Robin coefficient. Tangent-line continuations of powers of h_δ/b then give stability tests with a nonpositive boundary term and a bulk error controlled by the Hessian estimate. A Sobolev iteration gives the supremum bound.

Theorem 1.2 applies with the same constants for every e_R, so the vanishing blow-down errors force the direction-independent length defect $1 - |Du|^2$ to vanish at each fixed point. Harmonicity then forces $D^2 u = 0$. Section 5 classifies the resulting affine pieces.

Proposition 7.1 in Section 7 bounds the Hessian energy when the height excess is small. Appendix B gives the weighted local curvature estimate.

2. Stability on a spherical link

Fernández-Real's survey states the half-space classification through dimension six for nonzero one-homogeneous stable critical points of J whose positive phase lies locally on one side of a smooth free boundary away from the origin [7, Theorem 3 and Remark 4]. Jerison and Savin exclude nonflat stable cones that are classical off the origin in dimensions two through four [11, Theorem 1.1]. For dimensions five and six we verify the exclusion for classical stable cones by adapting the spherical tensor, flux, and cutoff arguments of [13, Sections 3, 5, and 7]. The pointwise flux comparisons follow from the local tensor identities and Bernoulli boundary data. We derive the angular stability inequality used to integrate them from deformations compactly supported away from the vertex. This is also the stability condition in the finite-index application.

Let u be a nonzero one-homogeneous Bernoulli solution in $\mathbb{R}^n$, $n \in \{5, 6\}$, that is classical and stable away from the origin. Write $u(r\theta) = r g(\theta)$ for $r > 0$, $\theta \in \mathbb{S}^m$, and $m = n - 1$. Let D be a connected component of $\{g > 0\}$. The radial direction is tangent to the conical free boundary and normal to the unit sphere. Thus the intersection is transverse. Intersecting a one-sided free-boundary chart with the sphere gives a smooth spherical boundary chart. In a sufficiently small such chart, the positive side is connected, so each point of ∂D borders D alone. Thus $\overline{D}$ is a compact smooth domain with boundary, and g is smooth up to it from inside. All link derivatives use the round metric. Write N for the inward unit normal to ∂D within the sphere.

For a Euclidean displacement $V = \alpha\theta + Y$ with Y tangent to the sphere, $D_V r = \alpha$ and $D_V \theta = Y/r$. Hence $D_V u = \alpha g + \langle \nabla g, Y \rangle$ and $Du = g\theta + \nabla g$. To differentiate again, use the ambient derivative of a tangent field Z on the unit sphere: $D_Y Z = \nabla_Y Z - \langle Y, Z \rangle \theta$. The radial part of $D_Y \nabla g$ cancels $(Yg)\theta$ from $D_Y(g\theta)$, leaving $\nabla_Y \nabla g + gY$. Put $T = \nabla^2 g + gI$, $B = \nabla T$, and $w = |T|$, identifying symmetric bilinear forms with endomorphisms by the metric. Then $D_V(Du) = r^{-1} TY$.

Taking the Euclidean trace gives $0 = r\Delta_{\mathbb{R}^n} u = \Delta g + mg$. On ∂D, $g = 0$ and the Bernoulli condition gives $|\nabla g| = 1$. The tangential derivatives along ∂D vanish and g is positive on the inward side. Thus

$$\Delta g = -mg \quad \text{in } D, \qquad g = 0, \quad \nabla g = N \quad \text{on } \partial D. \tag{2.1}$$

The boundary is nonempty. Otherwise D would be the whole sphere, where integrating $\Delta g = -mg$ would contradict $g > 0$. The Hessian calculation above, in a frame with one radial and m spherical directions, reads

$$D^2 u(r\theta) = r^{-1} \operatorname{diag}(0, T(\theta)). \tag{2.2}$$

Equation (2.1) gives $\operatorname{tr} T = 0$. Use an orthonormal frame at the point of computation, with indices from 1 to m and repeated indices summed. Our curvature convention is $R(X, Y)Z = \langle Y, Z \rangle X - \langle X, Z \rangle Y$ on the unit sphere. The commutator of covariant derivatives acts on each covariant slot by minus the action of R on vectors. For $g_k = \nabla_k g$ and $g_{jk} = \nabla_j g_k$, this gives

$$\nabla_i g_{jk} - \nabla_j g_{ik} = -g_i \delta_{jk} + g_j \delta_{ik}.$$

The derivatives of the gI term cancel the right-hand side, so $B_{ijk} = \nabla_i T_{jk}$ is symmetric in i, j. Since T is symmetric, B is fully symmetric. Also $\sum_j B_{ijj} = \nabla_i \operatorname{tr} T = 0$, so the trace on any two slots vanishes. For T, the commutator in directions ℓ, i contributes $-T(R(e_\ell, e_i)e_j, e_k) - T(e_j, R(e_\ell, e_i)e_k)$. Substituting the curvature formula gives

$$\nabla_\ell B_{ijk} - \nabla_i B_{\ell jk} = \delta_{\ell j} T_{ik} + \delta_{\ell k} T_{ij} - \delta_{ij} T_{\ell k} - \delta_{ik} T_{\ell j}. \tag{2.3}$$

To contract this identity, set $j = \ell$ and sum over ℓ. On the left, $\sum_\ell \nabla_\ell B_{i\ell k} = \Delta T_{ik}$ by symmetry of B, while $\nabla_i \sum_\ell B_{\ell\ell k} = 0$. The right side is $mT_{ik} + T_{ik} - T_{ik} - \delta_{ik} \operatorname{tr} T$. This proves $\Delta T = mT$ for the rough Laplacian, the trace of the second covariant derivative. The product rule gives $\Delta(w^2) = 2\langle T, \Delta T\rangle + 2|B|^2$. Where $w > 0$, Cauchy–Schwarz applied to $w\nabla_\ell w = T_{ij} B_{\ell ij}$ bounds $|\nabla w|$. Thus

$$\Delta w^2 = 2mw^2 + 2|B|^2, \qquad |\nabla w| \le |B| \quad (w > 0). \tag{2.4}$$

Set $\sigma = g^2 + |\nabla g|^2$. Since $\nabla_i \sigma = 2gg_i + 2g_{ij}g_j = 2T_{ij}g_j$, we have

$$\Delta\sigma = 2(\nabla_i T_{ij})g_j + 2T_{ij}g_{ij} = 2T_{ij}(T_{ij} - g\delta_{ij}) = 2w^2.$$

Here $\nabla_i T_{ij} = B_{iij} = 0$ by trace-freeness of B. On ∂D, $\sigma = 1$ by (2.1). If $T \not\equiv 0$ on D, the strong maximum principle gives $\sigma < 1$ in D, and the Hopf lemma gives $\partial_N \sigma < 0$ at each boundary point. For X tangent to ∂D, differentiating $\nabla g = N$ along ∂D gives $TX = \nabla_X N$, because $g = 0$ there. Since $\nabla_X N$ is tangent and T is symmetric, TN is parallel to N. The tangential block of T is the second fundamental form with respect to N, with trace H_D. Since T has trace zero,

$$TN = -H_D N, \qquad \partial_N \sigma = -2H_D.$$

Consequently $w \ge H_D > 0$ on ∂D. The boundary flux uses the entries of B with an odd number of normal slots. For tangent vectors X, Y to ∂D, differentiate $T(N, Y) = 0$ along X. Using $\nabla_X N = TX$ and $\langle \nabla_X Y, N\rangle = -\langle Y, \nabla_X N\rangle = -T(X, Y)$, we get

$$0 = B(X, N, Y) + T(TX, Y) - T(N, N)T(X, Y).$$

By symmetry of B, this gives the first of the two identities

$$B(N, X, Y) = T(N, N)T(X, Y) - (T^2)(X, Y), \qquad B(N, N, N) = w^2.$$

Indeed, summing the first over an orthonormal tangent basis of ∂D gives $-T(N, N)^2 - \sum_{\alpha,\beta} T_{\alpha\beta}^2 = -w^2$. We used $\operatorname{tr} T = 0$ and the vanishing of the mixed entries of T. The second identity follows by taking the full trace of $B(N, \cdot, \cdot)$.

Let φ be smooth on $\overline{D}$, with support away from any singular link points when using this calculation on a regular portion of a link. At each regular boundary point of its support, a smooth boundary chart contains only the one positive side belonging to D. Finitely many such charts cover the compact boundary part of the support. Extend φ across these charts and use a partition of unity with support in the charts and the interior. The resulting ambient angular test agrees with φ on $\overline{D}$ and vanishes near every other positive component and every singular point in the ambient link. For $\zeta \in C_c^\infty((0, \infty))$, the product test $\zeta(r)\varphi(\theta)$ using this extension has support in a compact annulus on the regular part of the ambient cone.

On the conical side the tangent space is spanned by θ and the tangents to ∂D. The inward normal is $N(r\theta) = N(\theta)$. Its radial derivative is zero, and in each unit tangent direction Y along ∂D its Euclidean derivative is $r^{-1}\nabla_Y N$. There is no radial term in this derivative because $\langle Y, N\rangle = 0$. Thus the Euclidean mean curvature is $H(r\theta) = r^{-1}H_D(\theta)$. The angular tangent directions scale by r, so the bulk measure is $r^{n-1}\,dr\,d\theta$ and the side measure is $r^{n-2}\,dr\,dS_\theta$. For the product test,

$$|D(\zeta\varphi)|^2 = |\zeta'|^2\varphi^2 + r^{-2}\zeta^2|\nabla\varphi|^2.$$

Substituting into (1.2) gives a common radial factor $\int_0^\infty r^{n-3}\zeta^2\,dr$ in the boundary and angular terms. Dividing by it for $\zeta \neq 0$ yields

$$\int_{\partial D} H_D\varphi^2 \leq \int_D |\nabla\varphi|^2 + \frac{\int_0^\infty r^{n-1}|\zeta'|^2\,dr}{\int_0^\infty r^{n-3}\zeta^2\,dr}\int_D \varphi^2.$$

Put $\mu = (n-2)/2$ and choose $\zeta(r) = r^{-\mu}\chi(\log r/L)$, where χ is a nonzero smooth compactly supported function. Differentiating gives $\zeta'(r) = r^{-\mu-1}(L^{-1}\chi'(\log r/L) - \mu\chi(\log r/L))$. In both integrals the remaining radial factor is $r^{-1}dr = dt$ for $t = \log r$. The denominator is $\int_{\mathbb{R}} \chi(t/L)^2\,dt$ and the numerator is

$$\int_{\mathbb{R}} \left(L^{-1}\chi'(t/L) - \mu\chi(t/L)\right)^2\,dt.$$

The cross term vanishes by $\int_{\mathbb{R}} \chi'\chi = 0$, so the quotient is $\mu^2 + L^{-2}\int|\chi'|^2/\int\chi^2$. As $L \to \infty$ this gives

$$\int_{\partial D} H_D\varphi^2 \leq \int_D (|\nabla\varphi|^2 + \tfrac{(n-2)^2}{4}\varphi^2). \tag{2.5}$$

The angular support stays fixed in this limit. For a suspended link, we apply (2.5) to each fixed test away from the tips before removing its angular cutoffs.

2.1. **The flux inequality on a five-dimensional link.** In dimension six, (2.5) bounds the boundary integral of $H_D\varphi^2$ by the integral of $|\nabla\varphi|^2 + 4\varphi^2$. We look for a test h_{s} and a vector field Z whose divergence controls this same bulk expression with a strict positive term, while its outward flux is bounded above by $H_D h_{\mathrm{s}}^2$. Integrating would then contradict stability. The pointwise comparisons of [13, Propositions 5.2 and 5.4] come from an algebraic certificate [13, Proposition 4.1]. We verify the geometric inputs to those comparisons before applying stability to the cutoff tests.

On the part of a link in $\mathbb{S}^5$ where $w > 0$, set $a = 29/50$. In a spherical orthonormal frame put

$$A = T/w, \qquad X = B/w, \qquad p = \nabla g, \qquad q_l = A_{ij}X_{lij}.$$

The matrix A is symmetric and trace free with $|A| = 1$. The tensor X is symmetric and trace free, and $g^2 + |p|^2 \leq 1$. These are the tensor and coefficient constraints in [13, Proposition 4.1]. We use the geometric value $w > 0$ and the specialization $z = 1$. The auxiliary vector in the certificate is unrestricted. Differentiating $w^2 = T_{ij}T_{ij}$ and using the quotient rule for A gives the first two identities below. The last two use $p = \nabla g$ and $T = \nabla^2 g + gI$.

$$\nabla_l w = wq_l, \qquad \nabla_l A_{ij} = X_{lij} - q_l A_{ij}, \qquad \nabla_l p_i = wA_{li} - g\delta_{li}, \qquad \nabla_l g = p_l.$$

In a frame with vanishing connection coefficients at the point, the last three identities give the differentiation rules D_l for the coefficient variables at $z = 1$ [13, (4.2) and Lemma 5.1]. The product rule and $\nabla_l\delta_{ij} = 0$ extend this identification to the coefficient polynomials in A, p, g, δ. The factors w and X are differentiated separately.

Use the polynomials f, U, K, M, P from [13, Proposition 4.1]. The monomial codes and integer coefficient numerators are listed in [13, Appendix A.2, Table I]. Use the decoding rules and slot orders in [13, Section 6.1]. Sum the decoded monomials with each numerator divided by $Q = 250000$, as in [13, (4.1) and (6.1)]. With $C_{lijk} = K_{lijk} - K_{iljk}$ and $G_{lijk} = \delta_{li}U_{jk} + C_{lijk}$, the fields from [13, (5.1) and (5.6)] are

$$h_{\mathrm{s}} = w^a\sqrt{f}, \qquad Z_l = w^{2a}(G_{lijk}X_{ijk} + wM_l + P_l). \tag{2.6}$$

Orthogonal equivariance makes h_{s} and Z independent of the chosen frame. The certificate gives $f \geq c_* > 0$ on the admissible data. The interior comparison in [13, Proposition 5.2] uses $\Delta T = 5T$ and the skew commutator (2.3) to eliminate fourth derivatives of g. Indeed, the $\delta_{li}U_{jk}$ part of G contracts $w^{-1}\nabla_l B_{ijk}$ to $5U_{jk}A_{jk}$, while the skew part C_{lijk} only sees $(\nabla_l B_{ijk} - \nabla_i B_{ljk})/(2w)$.

In (2.3) the last two terms then give the same contraction as the first two, by interchanging l, i. Consequently

$$w^{-1}G_{lijk}\nabla_l B_{ijk} = 5U_{jk}A_{jk} + C_{lijk}(\delta_{lj}A_{ik} + \delta_{lk}A_{ij}),$$

as in [13, Lemma 5.1]. To see how this enters the certificate, differentiate Z and use $\nabla_l X_{ijk} = w^{-1}\nabla_l B_{ijk} - q_l X_{ijk}$. The product rule gives

$$\begin{aligned} w^{-2a}\operatorname{div} Z = {} & w^{-1}G_{lijk}\nabla_l B_{ijk} + (D_l G_{lijk} + (2a-1)q_l G_{lijk})X_{ijk} \\ & + w(D_l M_l + (2a+1)q_l M_l) + D_l P_l + 2aq_l P_l. \end{aligned}$$

Here D_l acts on the coefficient polynomials at $z = 1$. In the interior certificate [13, (4.3)], choose the free vector $\ell = \nabla h_{\mathrm{s}}/h_{\mathrm{s}}$, which is defined since $w > 0$ and $f > 0$. Differentiating $h_{\mathrm{s}}^2 = w^{2a}f$ gives $D_l f + 2aq_l f = 2f\ell_l$. The two terms involving ℓ on the left of the certificate therefore sum to $-f|\ell|^2 = -w^{-2a}|\nabla h_{\mathrm{s}}|^2$. Using the fourth-derivative contraction above, the full left-hand side is $w^{-2a}(\operatorname{div} Z - 4h_{\mathrm{s}}^2 - |\nabla h_{\mathrm{s}}|^2)$. The certificate bounds it below by $\epsilon_*(1+|\ell|^2)$ for a numerical $\epsilon_* > 0$. This gives the strict interior comparison with $c = \epsilon_*$.

On a regular side boundary, $g = 0$, $p = N$, and $Ap = \lambda p$ with $|p| = 1$ and $\lambda = -H_D/w < 0$. The boundary identities above give

$$\frac{B(N, Y_1, Y_2)}{w^2} = (\lambda A - A^2)(Y_1, Y_2), \qquad \frac{B(N, N, N)}{w^2} = 1$$

for tangent Y_1, Y_2. By [13, Lemma 5.3], the contraction of $N_l G_{lijk}$ with B/w^2 uses exactly these components with an odd number of normal slots. The coefficient P vanishes at $g = 0$. Substituting $X = w(B/w^2)$ and $H_D = -\lambda w$ gives

$$N \cdot Z + H_D h_{\mathrm{s}}^2 = w^{2a+1}\left(N_l G_{lijk}\frac{B_{ijk}}{w^2} + N_l M_l - \lambda f\right).$$

The bracket is the boundary polynomial in [13, (4.4)] after the odd-normal contraction. It is positive by the boundary certificate. The inward-normal sign agrees with [13, Proposition 5.4]. Both comparisons apply on an original link in $\mathbb{S}^5$ and on the regular part of a suspended link.

We have proved the comparisons

$$\operatorname{div} Z \geq |\nabla h_{\mathrm{s}}|^2 + 4h_{\mathrm{s}}^2 + cw^{2a}, \tag{2.7}$$

$$N \cdot Z + H_D h_{\mathrm{s}}^2 \geq 0 \quad \text{on the smooth side of the link.} \tag{2.8}$$

The normalized coefficient domain is compact, so the polynomials and their first derivatives in (A, p, g) are bounded there. Using $f \geq c_*$ and the displayed differentiation rules, [13, Lemma 5.5] gives

$$|h_{\mathrm{s}}| \leq Cw^a, \quad |\nabla h_{\mathrm{s}}| \leq Cw^a(1 + w + |B|/w), \quad |Z| \leq Cw^{2a}(1 + w + |B|/w). \tag{2.9}$$

Let $0 \leq \xi \leq 1$ be smooth up to the regular boundary and vanish near the zeros of w and any tips. All products below involving ξ or $\nabla\xi$ are extended by zero there. Integrate (2.7) against ξ^2. The divergence theorem for $\xi^2 Z$ uses outward normal $-N$. In the suspended case we may truncate the domain at $\tau = \varepsilon, \pi - \varepsilon$ inside the regions where $\xi = 0$, so the new boundary faces contribute zero. Using $-N \cdot Z \leq H_D h_{\mathrm{s}}^2$ gives

$$\begin{aligned} \int_D \xi^2(|\nabla h_{\mathrm{s}}|^2 + 4h_{\mathrm{s}}^2 + cw^{2a}) &\leq \int_D \xi^2 \operatorname{div} Z \\ &= -\int_{\partial D} \xi^2 N \cdot Z - 2\int_D \xi Z \cdot \nabla\xi \leq \int_{\partial D} H_D(\xi h_{\mathrm{s}})^2 - 2\int_D \xi Z \cdot \nabla\xi. \end{aligned}$$

Apply (2.5) to ξh_{s} with $n = 6$, so its mass coefficient is 4. The expansion

$$|\nabla(\xi h_{\mathrm{s}})|^2 = \xi^2|\nabla h_{\mathrm{s}}|^2 + h_{\mathrm{s}}^2|\nabla\xi|^2 + 2\xi h_{\mathrm{s}}\nabla h_{\mathrm{s}} \cdot \nabla\xi$$

allows us to cancel the common Dirichlet and mass terms before taking any limits. Thus

$$c\int_D \xi^2 w^{2a} \le \int_D \left(h_{\mathrm{s}}^2 |\nabla\xi|^2 + 2\xi(h_{\mathrm{s}}\nabla h_{\mathrm{s}} - Z)\cdot\nabla\xi \right). \tag{2.10}$$

Write $\mathcal{E}_\xi$ for the integrand on the right. Both dimensions use the same estimate from (2.9):

$$|\mathcal{E}_\xi| \le C w^{2a}|\nabla\xi|(|\nabla\xi| + 1 + w + |B|/w) \quad \text{on } \{w>0\}.$$

Differentiation of the cutoff near $w=0$ costs a factor $|B|/w$. The weight w^{2a} in (2.10) therefore leads to the integral of $w^{2a-2}|B|^2$. On an original smooth link in either dimension, use the regularization $W_\delta = w^2+\delta$ as in [13, Lemma 7.1]. It is smooth even where $w=0$. Equation (2.4) gives $\Delta W_\delta \ge 2|B|^2$. Also $\nabla_l W_\delta = 2T_{ij}B_{lij}$, so $|\nabla W_\delta|^2 \le 4w^2|B|^2 \le 4W_\delta|B|^2$. Because $a-1<0$, we use the upper bound for the gradient term in the chain rule. The lower bound is positive since $a>1/2$. Thus

$$\Delta W_\delta^a = aW_\delta^{a-1}\Delta W_\delta + a(a-1)W_\delta^{a-2}|\nabla W_\delta|^2 \ge 2a(2a-1)W_\delta^{a-1}|B|^2.$$

If $T\not\equiv 0$, then $w>0$ on ∂D. The zero set is therefore compact in D. Choose $0\le\psi\le 1$ in $C_c^\infty(D)$ equal to one near this set. Then

$$2a(2a-1)\int_D \psi W_\delta^{a-1}|B|^2 \le \int_D \psi\,\Delta W_\delta^a = \int_D W_\delta^a \Delta\psi \le \|W_\delta^a\|_\infty \int_D |\Delta\psi|.$$

The right side is bounded uniformly for $0<\delta\le 1$ by boundedness of w on $\overline{D}$. On $\{w>0\}$, the weight W_δ^{a-1} increases to w^{2a-2} as $\delta\downarrow 0$. Away from the neighborhood of the zero set, w is bounded away from zero on the compact closure. Monotone convergence therefore gives

$$\int_{\{w>0\}} w^{2a-2}|B|^2 < \infty. \tag{2.11}$$

In particular the function

$$\mathcal{M} = w^{2a}(1+|B|/w)^2 \quad \text{on } \{w>0\}, \qquad \mathcal{M}=0 \quad \text{on } \{w=0\},$$

is integrable on any original nonflat link, since $\mathcal{M}\le 2w^{2a}+2w^{2a-2}|B|^2$ on $\{w>0\}$. Fix a smooth nondecreasing cutoff ϑ with values in $[0,1]$, equal to zero on $[0,1]$ and one on $[2,\infty)$. The cutoff $\vartheta(jw)$ is smooth across the zero set because it vanishes on a neighborhood of it. On $\{w>0\}$,

$$|\nabla\vartheta(jw)| \le C|B|/w,$$

since the derivative is supported in $1\le jw\le 2$, where $j\le 2/w$ and $|\nabla w|\le|B|$. For an original link of a cone in $\mathbb{R}^6$, use $\xi_j=\vartheta(jw)$. The error $|\mathcal{E}_{\xi_j}|$ is bounded by $C\mathcal{M}$ because w is bounded. The suspended link requires an additional cutoff at each tip.

2.2. **The five-dimensional cone.** Suppose a stable cone in $\mathbb{R}^5$ has a nonflat component $D_0\subset\mathbb{S}^4$, with fields g_0, T_0, B_0, w_0 as above. Extend the cone by $\widetilde{u}(x,t)=u_0(x)$ on $\mathbb{R}^5\times\mathbb{R}$. The extension is one-homogeneous and solves the classical Bernoulli problem off the axis $\{x=0\}$, because $\Delta\widetilde{u}=\Delta u_0$ and $D\widetilde{u}=(Du_0,0)$. The product boundary has the original principal curvatures and one additional zero. For an ambient test ϕ supported in a compact set off the axis, each slice $\phi(\cdot,t)$ has compact support off the vertex of u_0. The product splitting of the Dirichlet integral and Fubini give

$$Q_{\widetilde{u}}(\phi) = \int_{\mathbb{R}} Q_{u_0}(\phi(\cdot,t))\,dt + \int_{\{\widetilde{u}>0\}} |\partial_t\phi|^2 \ge 0.$$

This is the stability needed on the regular part of the suspension.

The new link is the suspension in Figure 2. In coordinates it is given by

$$\theta = (\sin\tau\, y, \cos\tau), \qquad 0<\tau<\pi, \quad y\in D_0, \qquad g(\theta)=\sin\tau\, g_0(y).$$

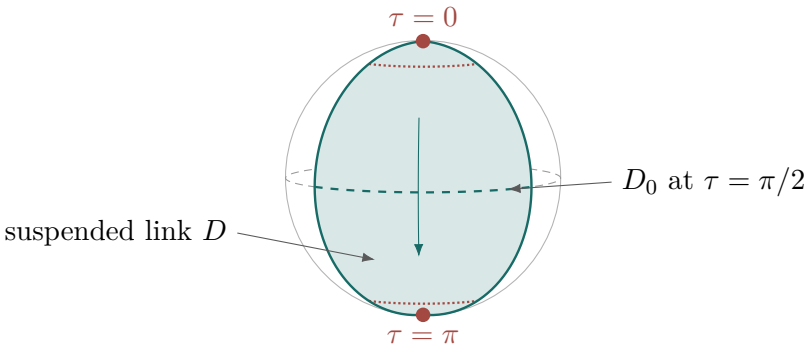


FIGURE 2. A lower-dimensional sketch of a spherical suspension. Each y in the base link runs along a meridian as τ varies. The dotted arcs mark the two latitudes where $\sin\tau = 1/j$ in the tip cutoff. The test vanishes in the caps beyond these arcs.

Write $s = \sin\tau$ and $c_\tau = \cos\tau$. If $E_1, \dots, E_4$ form an orthonormal tangent frame on the base sphere at y, then $e_\tau = (c_\tau y, -s)$ and $e_\alpha = (E_\alpha, 0)$ form an orthonormal frame at θ. Indeed, the derivative in a base direction E_α is $(sE_\alpha, 0)$, perpendicular to e_τ. The metric is $d\tau^2 + s^2 g_{\mathbb{S}^4}$, and the four scaled directions give $dV_D = s^4\, d\tau\, dV_{D_0}$.

The suspension formulas of [13, Proposition 3.4] can be seen directly from the ambient derivatives. The Hessian $D^2\widetilde{u}(\theta)$ is $D^2u_0(sy)$ with an extra zero row and column. From (2.2) and homogeneity, $w = w_0/s$. The product has the mean curvature of the original cone at sy, namely H_{D_0}/s. At unit radius this is also the link curvature $H_D > 0$. Moreover $g^2 + |\nabla g|^2 = |D\widetilde{u}(\theta)|^2 = |Du_0(y)|^2 = g_0^2 + |\nabla_0 g_0|^2 \le 1$. These facts verify the pointwise admissibility and boundary signs without applying a maximum principle at the tips.

To compute B, first note that for any one-homogeneous v, the restriction of D^3v at unit radius to three spherical tangent slots is the covariant derivative of the link tensor T. Indeed, differentiating $D^2v(Y,Z) = T(Y,Z)$ along a spherical tangent and subtracting the derivatives of the tangent fields gives ∇T. The radial parts of the ambient derivatives of the tangent fields make no contribution because D^2v vanishes in a radial slot. For $v = u_0$, differentiating $D^2u_0(ry) = r^{-1}D^2u_0(y)$ along the ray gives $D^3u_0(y)[y,\cdot,\cdot] = -D^2u_0(y)$. Consequently the third derivative vanishes when at least two slots are radial. By homogeneity $D^3u_0(sy) = s^{-2}D^3u_0(y)$, and the third derivative of the product ignores the last ambient coordinate. Evaluating it in the frame above gives

$$B_{\alpha\beta\gamma} = s^{-2}(B_0)_{\alpha\beta\gamma}, \qquad B_{\tau\alpha\beta} = -c_\tau s^{-2}(T_0)_{\alpha\beta}, \qquad B_{\tau\tau\alpha} = B_{\tau\tau\tau} = 0.$$

A single meridional slot occurs in three positions. Thus

$$|B|^2 = s^{-4}(|B_0|^2 + 3c_\tau^2 w_0^2). \tag{2.12}$$

Use $\xi_j(\tau, y) = \alpha_j(y)\beta_j(\tau)$, where $\alpha_j = \vartheta(jw_0)$ and $\beta_j = \vartheta(js)$. For each j, the support is separated from the tips by $s \ge 1/j$. In (2.5) the radial support has $r \ge r_{\min} > 0$, so on the product support $|x| = rs \ge r_{\min}/j$. This keeps each Euclidean test off the singular axis for its fixed radial cutoff.

The metric gives

$$|\nabla\xi_j|^2 = \alpha_j^2|\beta_j'|^2 + s^{-2}\beta_j^2|\nabla_0\alpha_j|^2.$$

On the support of β_j', $j \le 2/s$, so $|\beta_j'| = |j\vartheta'(js)c_\tau| \le C/s$. For $w_0 > 0$, set $b_0 = |B_0|/w_0$. The bound for the original cutoff gives $|\nabla_0\alpha_j| \le Cb_0$, and hence $|\nabla\xi_j| \le C(1+b_0)/s$. Equation (2.12) gives $|B|/w = s^{-1}\sqrt{b_0^2 + 3c_\tau^2}$. Since $w = w_0/s$ and w_0 is bounded, the factor $1 + w + |B|/w$ is also

bounded by $C(1+b_0)/s$. Multiplying the error estimate by the volume factor gives

$$s^4|\mathcal{E}_{\xi_j}| \le Cs^{4-2a-2}w_0^{2a}(1+b_0)^2 = Cs^{2-2a}\mathcal{M}_0(y),$$

where $\mathcal{M}_0$ is $\mathcal{M}$ on the base link. This majorant is integrable with respect to $d\tau\, dV_{D_0}$. The base factor $\mathcal{M}_0$ is integrable by (2.11). The factor s^{2-2a} is integrable since s vanishes linearly at both ends and $2-2a = 21/25 > -1$.

We can now pass to the limit in (2.10) in both dimensions. At each point where $w > 0$, the cutoff is eventually one with zero gradient. The error is already zero where $w = 0$. The preceding integrable majorants therefore give $\int_D \mathcal{E}_{\xi_j} \to 0$. On the left, $0 \le \xi_j^2 w^{2a} \le w^{2a}$, and w^{2a} is integrable on the smooth link by boundedness. On the suspension its density in the product coordinates is $s^{4-2a}w_0^{2a}$ with $4-2a = 71/25 > -1$, so it too is integrable. Dominated convergence gives $c\int_D w^{2a}$ as the limit on the left. This is positive because a nonflat component has $w > 0$ on an open set. The two limits contradict (2.10).

Thus $T = 0$ on each positive link component D in dimensions five and six. The gradient formula and the Hessian calculation show that $\theta \mapsto Du(\theta)$ has zero derivative in every tangent direction in D, so it is a constant vector e there. Taking its scalar product with θ gives $g(\theta) = e\cdot\theta$, and (2.1) gives $|e| = 1$ on taking the boundary trace of Du. By continuity $g = 0$ on ∂D, so D has no boundary in the open hemisphere $\{e\cdot\theta > 0\}$. Since that hemisphere is connected, D fills it. Any two open hemispheres with normals e, f intersect at $(e+f)/|e+f|$ if $e+f \ne 0$. Opposite hemispheres would put positive phase on both sides of the same equator, contrary to classicality at every point of the equator. Hence there is one positive hemisphere and $u = (e\cdot x)_+$. Together with [11, Theorem 1.1], this excludes nonflat cones in dimensions two through six. Every candidate in [9, Definition 4.1] is nonzero because 0 belongs to its free boundary. It has the required one-sided regularity and is stable under deformations supported away from the vertex by restriction. Hence $6 < n_*$ for the blow-down proposition.

3. A scalar defect for the gradient

Throughout this section and the next, the ambient dimension is six. In the positive phase Ω, put $p = Du$, $B = D^2u$, $w = |B|$, and $\sigma = 1-|p|^2$. The local estimate assumes $|p| \le 1$ in the positive phase of the ball under study, so $0 \le \sigma \le 1$ there. For a global classical solution, [4, Lemma 3.2] supplies this hypothesis on every ball.

Harmonicity gives $\Delta p = D\Delta u = 0$ and $\Delta|p|^2 = 2p\cdot\Delta p + 2|Dp|^2 = 2w^2$. On Γ, use the boundary Hessian identity $BN = -HN$ from Appendix A. Since $p = N$ and $\nu = -N$, it gives $B\nu = Hp$. In particular, $\partial_\nu\sigma = -2p\cdot B\nu = -2H$. The bulk and boundary identities are

$$\Delta p = 0, \qquad \Delta\sigma = -2w^2, \qquad \partial_\nu p = Hp, \qquad \partial_\nu\sigma = -2H, \tag{3.1}$$

where the last two hold on Γ. The components of p are the Jacobi fields from translations. Also, for $x \in \Gamma$ the point $x - t\nu(x)$ lies in Ω for small $t > 0$. Since $\sigma(x) = 0$ and $\sigma(x - t\nu(x)) \ge 0$, the outward derivative satisfies

$$-2H(x) = \partial_\nu\sigma(x) = \lim_{t\downarrow 0}\frac{\sigma(x) - \sigma(x - t\nu(x))}{t} \le 0.$$

Thus $H \ge 0$ on the free boundary in the ball.

For the fixed unit direction e, let P_e project onto $e^\perp$. Suppressing the dependence on e, write the defect from Section 1.2 as the length of $F = (P_e p, \sigma/\sqrt{30})$. Set

$$h^2 = |F|^2 = |P_e p|^2 + \frac{\sigma^2}{30}, \qquad h_\delta = (h^2+\delta^2)^{1/2}, \qquad 0 < \delta \le 1. \tag{3.2}$$

Figure 3 shows both contributions to the defect. Regularizing the length makes h_δ smooth at $F = 0$. The derivative of $h_\delta = |(F,\delta)|$ in a coordinate direction is the projection of the derivative

of (F, δ) onto its unit direction. The coercivity calculation below uses the remaining squared derivatives and the trace-free condition $\operatorname{tr} B = 0$.

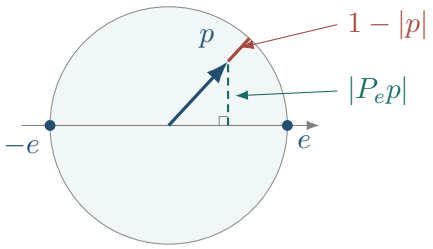


FIGURE 3. A planar slice of the unit ball of gradient values. The dashed segment measures $|P_e p|$. The radial gap measures $1 - |p|$, with $\sigma = (1 - |p|)(1 + |p|)$.

Lemma 3.1. *The regularized defect satisfies*

$$h_\delta \Delta h_\delta \geq \frac{w^2}{75}, \qquad |Dh_\delta|^2 \leq \frac{17}{15} w^2 \quad \text{in } \Omega, \qquad \partial_\nu h_\delta = H\left(h_\delta - \frac{\delta^2}{h_\delta}\right) \quad \text{on } \Gamma.$$

Proof. Differentiating the defect. Since P_e is constant, $D(P_e p) = P_e B$ and $\Delta(P_e p) = 0$. Also $D\sigma = -2Bp$ by symmetry of B. Thus

$$|DF|^2 = |P_e B|^2 + \frac{2}{15}|Bp|^2, \qquad F \cdot \Delta F = \frac{\sigma \Delta \sigma}{30} = -\frac{\sigma}{15} w^2.$$

Differentiating $h_\delta^2 = |F|^2 + \delta^2$ once and twice yields

$$h_\delta Dh_\delta = (DF)^{\mathsf{T}} F = B(P_e p) - \frac{\sigma}{15} Bp,$$
$$h_\delta \Delta h_\delta = |DF|^2 - |Dh_\delta|^2 - \frac{\sigma}{15} w^2.$$

Since $|F| \leq h_\delta$, Cauchy–Schwarz in the first identity gives $h_\delta^2 |Dh_\delta|^2 \leq |DF|^2 |F|^2$, where $|DF|^2$ is the sum of the squares of its entries. Dividing by h_δ^2 and using $|P_e B| \leq w$ and $|Bp| \leq w|p| \leq w$ gives $|Dh_\delta|^2 \leq (17/15) w^2$.

If $h = 0$, then $p = \pm e$, $\sigma = 0$, and $Dh_\delta = 0$. In that case the Laplacian identity gives

$$h_\delta \Delta h_\delta = |P_e B|^2 + \tfrac{2}{15}|Be|^2 \geq \tfrac{2}{15} w^2,$$

because $|P_e B|^2 + |Be|^2 = w^2$ by orthogonal projection and symmetry of B.

Recovering the Hessian from the remaining derivatives. At a point with $h > 0$, put $s = |P_e p|$ and choose an orthonormal basis $a, \xi_1, \dots, \xi_4$ of $e^\perp$ with $P_e p = sa$. If $s = 0$, any such basis works. Symmetry gives $|P_e B|^2 = |BP_e|^2 = |Ba|^2 + \sum_{j=1}^4 |B\xi_j|^2$. Set $U = Ba$, $V = 2Bp/\sqrt{30}$, $\alpha = s$, and $\beta = \sigma/\sqrt{30}$. In these variables,

$$|DF|^2 = \sum_{j=1}^4 |B\xi_j|^2 + |U|^2 + |V|^2, \qquad Dh_\delta = \frac{\alpha U - \beta V}{h_\delta}.$$

Since $h_\delta^2 \geq h^2 = \alpha^2 + \beta^2 > 0$, the subtraction of $|Dh_\delta|^2$ leaves at least $|U|^2 + |V|^2 - |\alpha U - \beta V|^2/(\alpha^2 + \beta^2)$ in addition to the four terms $|B\xi_j|^2$. Expanding the squares gives

$$(\alpha^2 + \beta^2)(|U|^2 + |V|^2) - |\alpha U - \beta V|^2 = |\beta U + \alpha V|^2.$$

It follows that

$$|DF|^2 - |Dh_\delta|^2 \geq \sum_{j=1}^4 |B\xi_j|^2 + \frac{|\sigma Ba + 2sBp|^2}{30h^2}.$$

Because $a \cdot p = s$, the squared norm of $\sigma a + 2sp$ is $\sigma^2 + 4\sigma s^2 + 4s^2(1-\sigma) = \sigma^2 + 4s^2$. It is positive because $h > 0$ means that s and σ cannot both vanish. Normalize this vector to $v = (\sigma a + 2sp)/|\sigma a + 2sp|$. Since $p = sa + (p \cdot e)e$, the vector v is orthogonal to the four ξ_j. The remaining term is a weight times $|Bv|^2$, with weight $|\sigma a + 2sp|^2/(30h^2)$. Thus

$$h_\delta \Delta h_\delta \geq E - \frac{\sigma}{15} w^2, \qquad E = \lambda |Bv|^2 + \sum_{j=1}^{4} |B\xi_j|^2, \qquad \lambda = \frac{\sigma^2 + 4s^2}{\sigma^2 + 30s^2} \in (0, 1].$$

Complete these five directions to an orthonormal basis with v at index 0, the four ξ_j at indices 1 through 4, and the remaining direction at index 5. Write $b_i = B_{ii}$ in this basis. The full squared Hessian norm is $w^2 = \sum_{i=0}^{5} b_i^2 + 2\sum_{i<j} B_{ij}^2$, while $E = E_{\mathrm{diag}} + E_{\mathrm{off}}$ with

$$E_{\mathrm{diag}} = \lambda b_0^2 + \sum_{i=1}^{4} b_i^2, \qquad E_{\mathrm{off}} = \sum_{0 \leq i < j \leq 5} (\omega_i + \omega_j) B_{ij}^2, \qquad \omega = (\lambda, 1, 1, 1, 1, 0).$$

The weights add because $B_{ij} = B_{ji}$. For every $i < j$ at least one of ω_i, ω_j is at least λ, so $2\sum_{i<j} B_{ij}^2 \leq (2/\lambda) E_{\mathrm{off}}$. The diagonal entry b_5 is controlled by $\operatorname{tr} B = 0$. Indeed, Cauchy–Schwarz with weights ω_i for $i \leq 4$ gives

$$b_5^2 = \left(\sum_{i=0}^{4} b_i \right)^2 \leq (1/\lambda + 4) E_{\mathrm{diag}}, \qquad \sum_{i=0}^{4} b_i^2 \leq E_{\mathrm{diag}}/\lambda.$$

The full diagonal sum is therefore at most $(2/\lambda + 4) E_{\mathrm{diag}}$. Adding the off-diagonal bound gives $w^2 \leq (2/\lambda + 4) E$, or $E \geq \lambda w^2/(2 + 4\lambda)$.

To absorb the term $-\sigma w^2/15$, use $s^2 \leq 1 - \sigma$. As a function of $r = s^2$, the fraction

$$\frac{\lambda}{2 + 4\lambda} = \frac{\sigma^2 + 4r}{6\sigma^2 + 76r}$$

has derivative $-52\sigma^2/(6\sigma^2 + 76r)^2$ wherever the denominator is positive. Thus it is bounded below by its value at $r = 1 - \sigma$. If $\sigma = 0$, then $h > 0$ implies $r > 0$ and the fraction is constant for $r > 0$. To compare the resulting lower bound minus $\sigma/15$ with $1/75$, multiply the difference by the positive factor $75(6\sigma^2 + 76(1-\sigma))$. The result is

$$\begin{aligned} &75(\sigma^2 + 4(1-\sigma)) - (5\sigma + 1)(6\sigma^2 + 76(1-\sigma)) \\ &\quad = 224 - 604\sigma + 449\sigma^2 - 30\sigma^3 \\ &\quad \geq 224 - 604\sigma + 419\sigma^2 = 419 \left(\sigma - \frac{302}{419} \right)^2 + \frac{2652}{419} > 0. \end{aligned}$$

The inequality uses $0 \leq \sigma \leq 1$. Consequently $E - \sigma w^2/15 \geq w^2/75$, as required.

The outward Robin identity. On Γ, $\sigma = 0$, $\partial_\nu(P_e p) = H P_e p$, and $H \geq 0$. In the normal derivative of h_δ^2, the length-defect term contributes $2\sigma \partial_\nu \sigma/30 = 0$. Using the remaining term gives

$$\partial_\nu h_\delta = \frac{F \cdot \partial_\nu F}{h_\delta} = \frac{H |P_e p|^2}{h_\delta} = H \left(h_\delta - \frac{\delta^2}{h_\delta} \right) \leq H h_\delta. \qquad \square$$

4. Stability and an iteration

The iteration uses the stability inequality to estimate powers of the defect on smaller balls. We work under the bound $|p| = |Du| \leq 1$ from Section 3 in the positive phase of the ambient ball. The free-boundary term has the right sign for a function satisfying a Robin inequality. We first derive the energy and Sobolev bounds, then construct powers that preserve this sign.

4.1. **Energy and Sobolev inequalities.** Let $G \geq 0$ be locally $C^{1,1}$ up to the free boundary and satisfy $\partial_\nu G \leq HG$ on Γ. Let η be smooth and compactly supported in the ambient ball. To use $\phi = \eta G$ in (1.2), choose a smooth partition of unity near its compact support on the closed positive side, subordinate to finitely many free-boundary charts and interior patches. Flatten each boundary chart and reflect the localized function across the flat face. Use chart cutoffs equal to one on the respective positive-side supports. The sum of the extensions then agrees with ϕ on the positive side and is Lipschitz with compact support in the ambient ball. Its mollifications ϕ_j converge uniformly and in H^1. Taking the mollification radii small keeps all supports in a fixed compact set K inside the ball. The bulk term in Q converges by H^1 convergence, and the absolute boundary error is bounded by

$$\|\phi_j^2 - \phi^2\|_{L^\infty(\Gamma\cap K)} \int_{\Gamma\cap K} |H|\, dS \longrightarrow 0.$$

The last integral is finite because the classical free boundary is smooth on this compact portion. Thus $Q(\eta G) \geq 0$.

The identity $|D(\eta G)|^2 = G^2|D\eta|^2 + DG \cdot D(\eta^2 G)$ follows by expanding both products. The locally Lipschitz field $X = \eta^2 G DG$ is supported in the support of ηG. Apply the divergence theorem to the pieces $\chi_i X$ using the same chart partition (χ_i). On the support of X we have $\sum_i \chi_i = 1$ and $\sum_i D\chi_i = 0$, so the interior terms involving $D\chi_i$ cancel. The field vanishes near the boundary of the ambient ball, so its only flux is through Γ, with the outward normal ν. This gives

$$\int_\Omega DG \cdot D(\eta^2 G) = -\int_\Omega \eta^2 G \Delta G + \int_\Gamma \eta^2 G \partial_\nu G.$$

Substituting into $Q(\eta G)$ yields

$$0 \leq Q(\eta G) = \int_\Omega G^2 |D\eta|^2 - \int_\Omega \eta^2 G \Delta G + \int_\Gamma \eta^2 G(\partial_\nu G - HG).$$

The last term is nonpositive. Here ΔG is the weak Laplacian, and $\partial_\nu G$ is taken from the positive side. We obtain

$$\int_\Omega \eta^2 G \Delta G \leq \int_\Omega G^2 |D\eta|^2. \tag{4.1}$$

For the Sobolev estimate, take f locally Lipschitz up to the free boundary with compact support in the ambient ball. Put $q = |f|^{5/2}$ on Ω and extend it by zero to $\widetilde{q}$ on $\mathbb{R}^6$. The Bernoulli condition bounds the jump of this extension. Indeed, $\operatorname{div}(qDu) = Du \cdot Dq$ in Ω, and its outward flux on Γ is $q\partial_\nu u = -q$. Therefore

$$\int_\Gamma q = -\int_\Omega Du \cdot Dq \leq \int_\Omega |Dq|.$$

For a smooth compactly supported vector field X, integrating $\operatorname{div}(qX) = Dq \cdot X + q \operatorname{div} X$ gives

$$-\int_{\mathbb{R}^6} \widetilde{q} \operatorname{div} X = \int_\Omega Dq \cdot X - \int_\Gamma q\, X \cdot \nu.$$

Thus $D\widetilde{q}$ has an interior part $Dq\, dx$ and a boundary part $-q\nu\, dS$, with total variation bounded by $\int_\Omega |Dq| + \int_\Gamma q \leq 2\int_\Omega |Dq|$.

For a nonnegative smooth mollifier ρ_ϵ of integral one, the derivative of $\widetilde{q} * \rho_\epsilon$ is $D\widetilde{q} * \rho_\epsilon$. Fubini's theorem and the unit mass of the mollifier give

$$\|D(\widetilde{q} * \rho_\epsilon)\|_{L^1(\mathbb{R}^6)} \leq \int_{\mathbb{R}^6} \int_{\mathbb{R}^6} \rho_\epsilon(x-y)\, d|D\widetilde{q}|(y)\, dx = |D\widetilde{q}|(\mathbb{R}^6).$$

Apply the Euclidean $W^{1,1}$ Sobolev inequality to the mollifications and use Fatou's lemma as they converge almost everywhere to $\widetilde{q}$. Since $(5/2)(6/5) = 3$ and $|Dq| \leq (5/2)|f|^{3/2}|Df|$, this gives

$$\|f\|_{L^3(\Omega)}^{5/2} = \|\widetilde{q}\|_{L^{6/5}(\mathbb{R}^6)} \leq C \int_\Omega |f|^{3/2}|Df| \leq C\|f\|_{L^3(\Omega)}^{3/2} \|Df\|_{L^2(\Omega)}.$$

The last step is Cauchy–Schwarz. Dividing when $\|f\|_{L^3(\Omega)} > 0$ proves the estimate

$$\|f\|_{L^3(\Omega)} \le C\|Df\|_{L^2(\Omega)}. \tag{4.2}$$

In particular, if $|DG|^2 \le AG\Delta G$ almost everywhere for some $A > 0$, then (4.1) gives

$$\int_\Omega |D(\eta G)|^2 \le 2\int_\Omega (\eta^2|DG|^2 + G^2|D\eta|^2) \le 2(A+1)\int_\Omega G^2|D\eta|^2.$$

Applying (4.2) to ηG gives

$$\|\eta G\|^2_{L^3(\Omega)} \le C(A+1)\int_\Omega G^2|D\eta|^2. \tag{4.3}$$

4.2. **Raising the power of the defect.** Near a point of Γ where $h > 0$, $h_\delta \to h$ in C^1. The Robin identity in Lemma 3.1 therefore gives $\partial_\nu h = Hh$, so $\partial_\nu(h^k) - Hh^k = (k-1)Hh^k$. For $k > 1$ this can be positive. To preserve the boundary inequality when taking powers, divide by a positive multiplier with the same Robin coefficient. The boundary identities $\sigma = 0$ and $\partial_\nu\sigma = -2H$ suggest $b = 1 - \sigma/2$, so set

$$b = \frac{1+|p|^2}{2}, \qquad t = \frac{h_\delta}{b}, \qquad G = bF(t). \tag{4.4}$$

We have $1/2 \le b \le 1$. Differentiation gives $Db = D^2u\,p$, so $|Db| \le w$ for $w = |D^2u|$. Equation (3.1) gives $\Delta b = w^2$. On Γ, $b = 1$ and $\partial_\nu b = H = Hb$. In the quotient rule the term $t\partial_\nu b = Hh_\delta$ cancels the Hh_δ term in Lemma 3.1, giving

$$\partial_\nu t = \frac{\partial_\nu h_\delta - t\partial_\nu b}{b} = \frac{H(h_\delta - \delta^2/h_\delta) - Hh_\delta}{b} = -\frac{H\delta^2}{h_\delta} \qquad \text{on } \Gamma.$$

For nonnegative C^1 functions F with $F' \ge 0$, the boundary inequality for $G = bF(t)$ follows from

$$\begin{aligned}\partial_\nu G - HG &= F(t)(\partial_\nu b - Hb) + bF'(t)\partial_\nu t \\ &= -HF'(t)\delta^2/h_\delta \le 0.\end{aligned} \tag{4.5}$$

Here we used $H \ge 0$ from Section 3.

For a twice differentiable F, the bulk product rule gives

$$\Delta G = F(t)\Delta b + F'(t)\big(b\Delta t + 2Db\cdot Dt\big) + bF''(t)|Dt|^2.$$

Since $bt = h_\delta$, the middle bracket equals $\Delta h_\delta - t\Delta b$. With $\Delta b = w^2$, this becomes

$$\Delta G = F'(t)\Delta h_\delta - (tF'(t) - F(t))w^2 + bF''(t)|Dt|^2. \tag{4.6}$$

The term $-(tF' - F)w^2$ must be absorbed by $F'\Delta h_\delta$. For a power $F(t) = t^k$, the ratio $(tF' - F)/F'$ is $(1 - 1/k)t$, so its size grows with t. Use the power up to $d = (150\sqrt{2})^{-1}$ and continue by its tangent line. For $k = 1$ or $k \ge 3/2$, define

$$F_k(t) = \begin{cases} t^k, & 0 \le t \le d, \\ d^k + kd^{k-1}(t-d), & t > d. \end{cases}$$

The continuation keeps F_k convex and holds the ratio at its value at d for $t \ge d$. For $t > 0$, set $T_d = \min(t, d)$. We use $G = bF_k(t)$ in stability and compare its norms to those of T_d^k to perform the iteration. Figure 4 illustrates the continuation and the clipped power for $k = 2$.

In terms of T_d, the two branches give

$$\begin{aligned} F_k(t) &= T_d^k + kT_d^{k-1}(t - T_d), & F_k'(t) &= kT_d^{k-1}, \\ tF_k'(t) - F_k(t) &= (k-1)T_d^k = (1-1/k)T_dF_k'(t). \end{aligned}$$

In particular, $0 \le tF_k' - F_k \le dF_k'$. Using $|P_e p| \le |p|$ and $0 \le \sigma \le 1$ gives $h^2 \le 1 - \sigma + \sigma^2/30 \le 1$. Thus $\delta \le h_\delta \le \sqrt{2}$ and, since $1/2 \le b \le 1$, $\delta \le t \le 2\sqrt{2}$. Thus F_k' is Lipschitz on the range of t

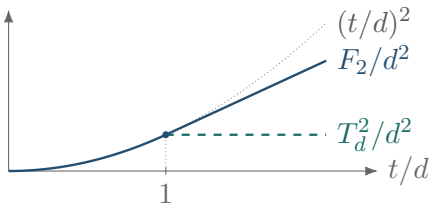


FIGURE 4. The power, its tangent-line continuation, and the clipped power for $k = 2$. All three coincide for $t \le d$. The continuation preserves the slope at d, while the clipped power stays constant above d.

for fixed $\delta > 0$, so $G = bF_k(t)$ is locally $C^{1,1}$ up to Γ. This regularity suffices to use the energy inequality for each fixed δ. The estimates below will be uniform in δ.

The product rule (4.6) holds almost everywhere for this G. At $t = d$ we may set $F_k''(d) = 0$. The implicit function theorem covers the points on that level with $Dt \neq 0$ by smooth hypersurface patches. A countable subcover suffices, so this part of the level set has zero volume. At a point x of the level with $Dt(x) = 0$, $t(y) - d = o(|y - x|)$. The Lipschitz bound for F_k' then gives $F_k'(t(y)) - F_k'(d) = o(|y - x|)$, so the derivative of $F_k'(t)$ at x is zero. This justifies the chain rule on the rest of the level. Local $C^{1,1}$ regularity ensures that the weak Laplacian is represented by this almost-everywhere formula.

By Lemma 3.1 and $h_\delta \le \sqrt{2}$, $\Delta h_\delta \ge w^2/(75h_\delta) \ge 2dw^2$. Since $0 \le tF_k' - F_k \le dF_k'$, at most half of the lower bound $2dF_k'w^2$ is used to absorb the negative term in (4.6). Thus

$$\Delta G \ge dF_k'w^2 + bF_k''|Dt|^2. \tag{4.7}$$

Multiplying by $G = bF_k$ gives the two nonnegative terms $d\,bF_kF_k'w^2$ and $b^2F_kF_k''|Dt|^2$ in the lower bound for $G\Delta G$. We use them to control $|DG|^2$. For $k = 1$, $F_1(t) = t$ and $G = h_\delta$. The two bulk bounds in Lemma 3.1 give $|DG|^2 \le 85G\Delta G$, since $75(17/15) = 85$.

For $k \ge 3/2$ and $t < d$, the derivatives of t^k give

$$(F_k')^2 = \frac{k}{k-1}F_kF_k'' \le 3F_kF_k'', \qquad F_k^2 = \frac{t}{k}F_kF_k' \le dF_kF_k'.$$

Differentiate $G = bF_k(t)$ and use these bounds to get

$$|DG|^2 \le 2b^2(F_k')^2|Dt|^2 + 2F_k^2w^2 \le 6b^2F_kF_k''|Dt|^2 + 4d\,bF_kF_k'w^2 \le 6G\Delta G.$$

The middle inequality uses $b \ge 1/2$, and the last uses (4.7). For $t \ge d$, differentiate $bt = h_\delta$ to get

$$DG = F_k'Dh_\delta - (tF_k' - F_k)Db.$$

The bounds on Dh_δ and Db yield $|DG| \le (\sqrt{17/15} + d)F_k'w$. Also $F_k' \le (k/d)F_k$ there, since $F_k' = kd^{k-1}$ and $F_k \ge d^k$. Thus

$$|DG|^2 \le (\sqrt{17/15} + d)^2(F_k')^2w^2 \le \frac{(\sqrt{17/15} + d)^2k}{d}F_kF_k'w^2 \le \frac{2(\sqrt{17/15} + d)^2k}{d^2}G\Delta G.$$

The last bound uses $G\Delta G \ge (d/2)F_kF_k'w^2$, since $b \ge 1/2$. Combining both branches and the case $k = 1$ gives

$$|DG|^2 \le C_dk\,G\Delta G \tag{4.8}$$

almost everywhere.

The formula for F_k gives $F_k/T_d^k = 1 + k(t - T_d)/T_d$. The second term is zero for $t \le d$. For $t > d$, $T_d = d$ and $t \le 2\sqrt{2}$, so $1 \le F_k/T_d^k \le 1 + 2\sqrt{2}k/d$ everywhere. With $1/2 \le b \le 1$, we have

$\frac{1}{2}T_d^k \le G \le C_d k T_d^k$. When the ambient ball is B_1, choose a cutoff $0 \le \eta \le 1$ that is one on B_r, supported in B_ρ, and satisfies $|D\eta| \le C/(\rho - r)$, where $0 < r < \rho \le 1$. Using (4.3) and (4.8),

$$\begin{aligned}\|T_d\|^{2k}_{L^{3k}(\Omega\cap B_r)} &\le \|\eta T_d^k\|^2_{L^3(\Omega)} \le 4\|\eta G\|^2_{L^3(\Omega)} \\ &\le C_d k \int_\Omega G^2 |D\eta|^2 \le \frac{C_d k^3}{(\rho - r)^2}\|T_d\|^{2k}_{L^{2k}(\Omega\cap B_\rho)}.\end{aligned}$$

The factor k^3 comes from k in (4.8) and k^2 in $G^2 \le C_d^2 k^2 T_d^{2k}$. Taking $2k$-th roots gives the iteration step

$$\|T_d\|_{L^{3k}(\Omega\cap B_r)} \le \left(\frac{C_d k^3}{(\rho - r)^2}\right)^{1/(2k)} \|T_d\|_{L^{2k}(\Omega\cap B_\rho)} \qquad (k = 1 \text{ or } k \ge 3/2). \tag{4.9}$$

Proof of Theorem 1.2. Set $y = x_0 + Rx$ and $u_R(x) = u(y)/R$ on B_1. The Bernoulli problem, gradient bound, and stability are preserved by (A.1). At corresponding points $Du_R(x) = Du(y)$, so both terms of the defect are preserved. If $\Omega_R = \{u_R > 0\}$ and $h_{e,R}$ is its defect, then

$$\int_{\Omega_R\cap B_1} h_{e,R}^2\, dx = R^{-6} \int_{\Omega\cap B_R(x_0)} h_e^2\, dy = E_e^2.$$

Use Ω and h for the rescaled positive phase and defect from now on.

Fix $0 < \delta \le 1$ and take $k_j = (3/2)^j$ and $r_j = 1/2 + 2^{-j-1}$ for $j \ge 0$. Then $k_0 = 1$, $r_0 = 1$, and $3k_j = 2k_{j+1}$. The exponent on the smaller ball in one application of (4.9) is the input exponent for the next. Applying it from r_j to r_{j+1} for $j = 0, \dots, m-1$ gives

$$\|T_d\|_{L^{2k_m}(\Omega\cap B_{r_m})} \le \left[\prod_{j=0}^{m-1} \left(\frac{C_d k_j^3}{(r_j - r_{j+1})^2}\right)^{1/(2k_j)}\right] \|T_d\|_{L^2(\Omega\cap B_1)}.$$

Since $r_j - r_{j+1} = 2^{-j-2}$, the logarithm of the factor at step j is

$$\frac{\log C_d + 3j\log(3/2) + (2j+4)\log 2}{2k_j}.$$

The sum of the absolute values of these logarithms is finite because $\sum_{j\ge0}(j+1)(2/3)^j < \infty$. The products are therefore bounded by a dimensional constant independent of m and δ. Restrict the norm on the left to $\Omega \cap B_{1/2}$ and use $T_d \le t = h_\delta/b \le 2h_\delta$ on the right. This gives, for every m,

$$\|T_d\|_{L^{2k_m}(\Omega\cap B_{1/2})} \le C\|h_\delta\|_{L^2(\Omega\cap B_1)}.$$

Write M_δ for the right-hand side. If $T_d > M_\delta$ at a point of $\Omega \cap B_{1/2}$, continuity gives a ball U about that point in this set and a number $a > M_\delta$ with $T_d \ge a$ on U. The last norm is then at least $a|U|^{1/(2k_m)}$, which tends to $a > M_\delta$ as $m \to \infty$. This contradicts the norm bound, so $T_d \le M_\delta$ pointwise.

Now let $\delta \downarrow 0$. The left side $T_d = \min(h_\delta/b, d)$ converges pointwise to $\min(h/b, d)$. On the right,

$$\|h_\delta\|^2_{L^2(\Omega\cap B_1)} = E_e^2 + \delta^2|\Omega \cap B_1| \longrightarrow E_e^2.$$

Thus $\min(h/b, d) \le CE_e$ on $\Omega \cap B_{1/2}$. The smallness hypothesis now removes the cutoff. Choose $\varepsilon_0 = d/(2C)$. If $E_e \le \varepsilon_0$, then $CE_e \le d/2 < d$, so $h/b < d$ at every point of this set. Consequently $h = b(h/b) \le bCE_e \le CE_e$. The coercivity bounds in Lemma 3.1, the Sobolev constant, and the numerical cutoff d are independent of the unit vector e. Thus C and ε_0 are uniform in e. The gradient is unchanged by the rescaling and $B_{1/2}$ corresponds to $B_{R/2}(x_0)$, which proves the stated estimate. □

5. Blow-down and classification

The direction supplied by the blow-down may vary with the radius. The length defect $1-|Du|^2$ is independent of that direction, so it can be estimated at a fixed point as the radius grows.

Proof of Theorem 1.1. If $\Gamma = \varnothing$, then Ω is both open and closed in $\mathbb{R}^6$. If $\Omega = \varnothing$, then $u = 0$. Otherwise $\Omega = \mathbb{R}^6$ by connectedness and u is nonnegative and harmonic everywhere. The ball mean-value formula for Du and the divergence theorem give, for each $x \in \mathbb{R}^6$ and $R > 0$,

$$Du(x) = \frac{1}{|B_R|}\int_{B_R(x)} Du(y)\,dy = \frac{1}{|B_R|}\int_{\partial B_R(x)} u(y)\nu_{B_R}(y)\,dS_y,$$

where ν_{B_R} is the outward normal to the sphere. Nonnegativity and the sphere mean-value formula give

$$|Du(x)| \le \frac{1}{|B_R|}\int_{\partial B_R(x)} u(y)\,dS_y = \frac{|\partial B_R|}{|B_R|}u(x) = \frac{6u(x)}{R}.$$

Letting $R \to \infty$ gives $Du(x) = 0$, so u is constant.

Normalization and blow-down. Assume $\Gamma \neq \varnothing$, and suppose that $D^2u(x_*) \neq 0$ at some $x_* \in \Omega$. Choose $x_0 \in \Gamma$. A single Bernoulli scaling, fixed throughout the contradiction argument, gives

$$v(x) = \frac{u(x_0 + Lx)}{L}, \qquad Dv(x) = Du(x_0 + Lx), \qquad D^2v(x) = LD^2u(x_0 + Lx).$$

Its positive phase is $\Omega_v = L^{-1}(\Omega - x_0)$, so $0 \in \partial\Omega_v$. The gradient formula preserves the Bernoulli condition, and the Hessian formula gives $\Delta v(x) = L\Delta u(x_0 + Lx) = 0$ in Ω_v. The translation and dilation preserve smoothness and the local positive side. By (A.1), the scalar form scales by L^{-4}. Appendix A therefore gives stationarity and inner stability for v, as required in [9, Definition 3.2].

Choose $L > \max\{|x_* - x_0|, |D^2u(x_*)|^{-1}\}$. Then $y_* = (x_* - x_0)/L$ lies in $\Omega_v \cap B_1$ and

$$|D^2v(y_*)| = L|D^2u(x_*)| > 1.$$

By continuity this strict bound holds on an open neighborhood of y_* in $\Omega_v \cap B_1$. This gives the essential supremum bound required in [9, (4.11)].

$$\|D^2v\|_{L^\infty(\Omega_v \cap B_1)} \ge 1.$$

Section 2 gives $6 < n_*$ for the critical dimension in [9, Definition 4.1]. We can thus apply [9, Proposition 4.12] to v. For each $R > 0$ it gives a unit vector e_R and a profile $V_R(x) = |e_R \cdot x|$. Squaring the gradient term of [9, (4.12)] gives

$$R^{-6}\int_{B_R} |Dv - DV_R|^2 \le \omega(R^{-1})^2 \longrightarrow 0 \qquad \text{as } R \to \infty. \tag{5.1}$$

Here ω is the modulus in the proposition and the gradients are weak gradients on the whole ball. Under $x = Ry$, the volume element is $dx = R^6\,dy$, which cancels R^{-6}. Since $D_y(v(Ry)/R) = Dv(Ry)$ and $DV_R(Ry) = DV_R(y)$ almost everywhere, the left-hand side of (5.1) is $\|D(v(R\cdot)/R) - DV_R\|^2_{L^2(B_1)}$.

From the averaged error to a pointwise identity. Since v is global and classical, [4, Lemma 3.2] gives $|Dv| \le 1$ in Ω_v in dimension six. To compare the two errors, take $|p| \le 1$, a unit vector e, and $q \in \{e, -e\}$. Since $P_eq = 0$ and $|q + p| \le 2$,

$$|P_ep| = |P_e(p - q)| \le |p - q|, \qquad 0 \le 1 - |p|^2 = (q - p)\cdot(q + p) \le 2|p - q|.$$

Squaring and using the coefficient $1/30$ in the definition of the defect gives the pointwise comparison

$$|P_ep|^2 + \frac{(1 - |p|^2)^2}{30} \le \frac{17}{15}|p - q|^2. \tag{5.2}$$

Off the measure-zero plane $\{e_R \cdot x = 0\}$, we have $DV_R(x) = \operatorname{sgn}(e_R \cdot x)e_R$. Apply (5.2) with $p = Dv(x)$ and $q = DV_R(x)$ on $\Omega_v \cap B_R$. Writing h_e for the defect of v, we obtain

$$\begin{aligned} E_R^2 := R^{-6} \int_{\Omega_v \cap B_R} h_{e_R}^2 &\le \frac{17}{15} R^{-6} \int_{\Omega_v \cap B_R} |Dv - DV_R|^2 \\ &\le \frac{17}{15} R^{-6} \int_{B_R} |Dv - DV_R|^2 \le \frac{17}{15}\, \omega(R^{-1})^2. \end{aligned}$$

Thus $E_R \to 0$, so $E_R \le \varepsilon_0$ for all sufficiently large R. The restriction of v to B_R is classical stable, since variations compactly supported in B_R are admissible globally. Apply Theorem 1.2 on B_R with the direction e_R chosen at that radius. Its uniform constants give $\sup_{\Omega_v \cap B_{R/2}} h_{e_R} \le C E_R$.

Fix $x \in \Omega_v$. For every sufficiently large $R > 2|x|$, the point x belongs to $B_{R/2}$, and hence

$$0 \le 1 - |Dv(x)|^2 \le \sqrt{30}\, h_{e_R}(x) \le \sqrt{30}\, C E_R.$$

The middle bound uses the squared length term in $h_{e_R}^2$. The quantity $1 - |Dv(x)|^2$ is the same for all R, even though e_R can vary. Letting $R \to \infty$ gives $|Dv(x)|^2 = 1$ for every $x \in \Omega_v$. Within the positive phase we can differentiate this identity. Since v is harmonic there, the product rule gives

$$0 = \Delta |Dv|^2 = 2 \sum_{i,j=1}^{6} (\partial_{ij} v)^2 + 2 \sum_{i=1}^{6} (\partial_i v) \partial_i (\Delta v) = 2 |D^2 v|^2.$$

This contradicts $|D^2 v(y_*)| > 1$. We conclude that $D^2 u = 0$ throughout the original positive phase Ω.

Classification of the positive components. Let U be a connected component of Ω. The equation $D^2 u = 0$ makes Du locally constant, hence constant on U, say $Du = e_U$. For any $x_U \in U$ we have

$$u(x) = \ell_U(x) := u(x_U) + e_U \cdot (x - x_U) \qquad (x \in U),$$

because $u - e_U \cdot x$ has zero gradient on the connected set U. Since $\Gamma \neq \varnothing$, U is a nonempty proper open subset of $\mathbb{R}^6$ and has nonempty boundary. If $z \in \partial U$ had $u(z) > 0$, a small connected ball about z in Ω would meet U and therefore lie in U, contradicting $z \in \partial U$. Thus $u(z) = 0$ and $z \in \Gamma$. Taking a sequence in U tending to z gives $\ell_U(z) = 0$ by continuity. The positive-side trace of Du there equals e_U, so the Bernoulli condition gives $|e_U| = 1$.

It follows that U fills the open half-space $\{\ell_U > 0\}$. Indeed, U is relatively open there, and a limit point of U in that half-space must belong to U, since $\partial U \subset \{\ell_U = 0\}$. The half-space is connected, so its nonempty relatively open and closed subset U is the whole half-space. Write $\ell_U(x) = e_U \cdot x - c_U$.

Consider distinct components $U_i = \{e_i \cdot x > c_i\}$, $i = 1, 2$. If $e_1 + e_2 \neq 0$, then for $x = t(e_1 + e_2)$ we have $e_i \cdot x = t(1 + e_1 \cdot e_2)$ for both i. The coefficient $1 + e_1 \cdot e_2 = \frac{1}{2}|e_1 + e_2|^2$ is positive, so for large $t > 0$ this common value exceeds both c_1 and c_2. This would put x in both components. Hence the normals of any two distinct components are opposite. A third component would have to have a normal opposite to both e_1 and e_2, which is impossible.

A single component gives $u = (e_U \cdot x - c_U)_+$. With two, write the components as $\{e \cdot x > b\}$ and $\{e \cdot x < a\}$. A point with $b < e \cdot x < a$ would lie in both, so disjointness gives $a \le b$. If $a = b$, the two half-spaces fill the complement of their common boundary plane, which is then the free boundary. For any point z on that plane and any $t > 0$, the points $z + te$ and $z - te$ lie in opposite positive components. Positivity on both sides of the plane violates the local one-sidedness of a classical solution. Thus $a < b$, and the zero phase is exactly the closed strip $\{a \le e \cdot x \le b\}$ of width $b - a > 0$. On the two components the affine values are $e \cdot x - b$ and $a - e \cdot x$, respectively, which gives the second profile in Theorem 1.1. Figure 1 shows both profiles. □

Proof of Corollary 1.3. For $1 \le n \le 5$, set $m = 6 - n$ and $U(x, y) = u(x)$ for $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$. Its positive phase and free boundary are $\Omega \times \mathbb{R}^m$ and $\Gamma \times \mathbb{R}^m$. The product boundary is smooth with a single positive side locally. In dimension one, each isolated free-boundary endpoint extends to a hyperplane, with the positive side given by the positive interval near that endpoint. The identities $\Delta U = \Delta u$ and $DU = (Du, 0)$ give harmonicity and preserve the boundary gradient length, so U is classical.

The inward unit normal is $(N(x), 0)$. The tangent directions split into the original tangent space and the added coordinate directions. The tangential divergence is $H(x)$, since the normal is constant in the added directions. The surface metric in product charts has the original surface metric and the identity as its two diagonal blocks, so the surface element is $dS_x\, dy$. For $n = 1$, dS_x is counting measure on the endpoints. The tangent space is zero-dimensional, so $H = 0$ and $Q_u(\psi) = \int_\Omega |\psi'|^2 \ge 0$. For $n \ge 2$, scalar stability of u follows from (1.2).

For $\phi \in C_c^\infty(\mathbb{R}^6)$, each slice $\phi_y = \phi(\cdot, y)$ has compact support in the x-projection of $\operatorname{supp} \phi$, so it is an admissible test for Q_u. Splitting $|D\phi|^2$ into $|D_x\phi|^2 + |D_y\phi|^2$ and using Fubini for the bulk and boundary integrals gives

$$Q_U(\phi) = \int_{\mathbb{R}^m} Q_u(\phi_y)\, dy + \int_{\Omega \times \mathbb{R}^m} |D_y\phi|^2\, dx\, dy \ge 0.$$

Appendix A now gives inner stability of U under arbitrary compactly supported deformations in $\mathbb{R}^6$.

Apply Theorem 1.1. Its Hessian conclusion gives $D^2u(x) = D_x^2U(x, 0) = 0$ in Ω. The free boundary of U is empty exactly when Γ is empty, and in this case the constant conclusion passes to u. Otherwise one of the two nonconstant profiles holds for U. On each positive component its gradient is e or $-e$. Since $D_yU = 0$, the unit vector e has the form $(e_0, 0)$ with $e_0 \in \mathbb{R}^n$ a unit vector. Restricting the profile to $y = 0$ gives the claimed formula for u. □

6. Local curvature and finite index

6.1. **Bounds on balls centered at the free boundary.** A ball containing a free-boundary point has a local bound on the gradient. The Bernoulli condition controls the solution near that point, and Harnack chains carry this control into the positive phase. Precisely, if v is classical in $B_2 \subset \mathbb{R}^n$, $n \ge 2$, and $0 \in \Gamma_v$, then

$$\sup_{B_{3/2}} v + \sup_{B_{3/2} \cap \{v > 0\}} |Dv| \le L_n. \tag{6.1}$$

Take $x \in B_{3/2} \cap \{v > 0\}$ with $d = \operatorname{dist}(x, \Gamma_v) \le 1/16$. A nearest point $p \in \Gamma_v$ exists, since the infimum can be taken in $\overline{B_{1/8}(x)} \subset B_2$. The connected ball $B_d(x)$ contains x and avoids Γ_v, so it is positive. Harnack's inequality gives $v \ge c_n v(x)$ on $\partial B_{d/2}(x)$. On the annulus, let $w(s)$ be the radial harmonic function with values $c_n v(x)$ at $s = d/2$ and zero at $s = d$, where s is distance from x. Thus

$$w(s) = \begin{cases} c_n v(x) \dfrac{s^{2-n} - d^{2-n}}{(d/2)^{2-n} - d^{2-n}}, & n > 2, \\ c_2 v(x) \dfrac{\log(d/s)}{\log 2}, & n = 2. \end{cases}$$

The boundary values give $v \ge w$ in the annulus by the maximum principle. Minimizing distance on the smooth free boundary makes $x - p$ orthogonal to its tangent space. Since the segment from p toward x is positive, $N(p) = (x - p)/d$. Along this segment $v \ge w$ and both vanish at p,

so $1 = \partial_N v(p) \geq -w'(d)$. Differentiation gives

$$-w'(d) = \begin{cases} \dfrac{c_n(n-2)}{2^{n-2}-1}\dfrac{v(x)}{d}, & n > 2, \\ \dfrac{c_2}{\log 2}\dfrac{v(x)}{d}, & n = 2. \end{cases}$$

Thus $v(x) \leq C_n d$. The interior gradient estimate and Harnack's inequality give

$$|Dv(x)| \leq \frac{C_n}{d} \sup_{B_{d/2}(x)} v \leq \frac{C_n}{d} v(x) \leq C_n.$$

If $d > 1/16$, the distance to Γ_v along the segment from x to 0 is continuous and ends at zero. Stop at the first point with distance $1/16$. The segment up to there stays in the positive phase, since an exit would require distance zero. Divide the segment into at most 96 equal steps of length at most $1/64$, since its length is at most $3/2$. Use the endpoints of these steps as centers. The balls of radius $1/32$ about them are positive and contained in B_2. Each next center is in the inner half of the preceding ball, so Harnack's inequality transfers the bound at the last center to x. Applying the interior gradient estimate and Harnack's inequality on $B_{1/32}(x)$ bounds $|Dv(x)|$. Together with $v = 0$ on the zero phase this proves (6.1).

The rescaling argument needs a Hessian bound on neighborhoods that grow to fill the whole space at the new scale. In taking the limit we must allow two free-boundary sheets to touch.

Let v_j be classical stable in B_{S_j}, with $S_j \to \infty$, $0 \in \Gamma_{v_j}$, and $|D^2 v_j| \leq M$ in the positive phase for a fixed $M > 0$. Pass to a subsequence with $S_j > j + 1$ and restrict to B_j to apply [9, Lemma 4.6]. The local Lipschitz hypothesis follows from classicality as in Section 1. The lemma gives a global 1-Lipschitz, inner stable limit v, with locally uniform and strong H^1_{loc} convergence. Both the closed positive phases and the free boundaries converge locally in Hausdorff distance. In particular $0 \in \Gamma_v$. On each ball compactly contained in $\{v > 0\}$ the v_j are eventually positive by locally uniform convergence. Interior harmonic estimates there give harmonicity of v and convergence of all derivatives on smaller balls.

The Hessian bound controls how quickly the gradient can turn along a segment in the positive phase. Fix a prelimit boundary point p in a bounded set and take j large enough that the rays below stay in B_{S_j}. If b is a unit vector with $b \cdot N(p) \geq 1/16$, the ray in direction b enters the positive phase. Its initial value is zero and its initial gradient is $N(p)$. Integrating the Hessian bound up to a possible first exit gives

$$v_j(p + sb) \geq s\, b \cdot N(p) - Ms^2/2, \qquad |Dv_j(p + sb) - N(p)| \leq Ms.$$

For $0 < s \leq \ell = 1/(16(1 + M))$ the lower bound is at least $s/32$. The inequalities hold up to a first exit by continuity, and the strictly positive value there rules out such an exit.

Take another boundary point q, with $0 < d = |p - q| \leq r_0 = 1/(10^4(1 + M))$. If $N(p) \cdot N(q) \geq -31/32$, set $b = (N(p) + N(q))/|N(p) + N(q)|$. Both scalar products with b equal $\sqrt{(1 + N(p) \cdot N(q))/2} \geq 1/8$. The point $p + 32db$ is within $33d < \ell$ of both p and q. Its displacement from q has scalar product at least $32d/8 - d = 3d$ with $N(q)$, so the corresponding unit direction has scalar product at least $3/33 > 1/16$. Both rays thus reach the same point through the positive phase. Comparing the two initial gradients with the gradient at this point gives $|N(p) - N(q)| \leq M(32d + 33d) = 65Md$. Since $d \leq r_0$,

$$N(p) \cdot N(q) = 1 - \tfrac{1}{2}|N(p) - N(q)|^2 \geq 1 - \tfrac{1}{2}(65/10^4)^2 > 31/32.$$

Including the case of nearly opposite normals, we obtain

$$|N(p) \cdot N(q)| \geq 31/32.$$

Use $N(p)$ as the vertical direction and set $r = r_0/16$. In the cylinder $|x'| < 4r$, $|x_n| < 4r$ about p, each boundary point has distance less than r_0 from p. Its normal has $|N_n| \geq 31/32$, so it lies on a local graph of slope $|N'|/|N_n| \leq \sqrt{63}/31 < 1/3$. There is at most one point of each orientation on any vertical line in this cylinder. Indeed, if two inward normals point up, the inward vertical ray from the lower point reaches the upper point within length ℓ while still in the positive phase. For two downward normals use the ray from the upper point. The same reasoning shows that when both orientations occur, the upward-facing graph lies above the downward-facing graph.

Each sheet meeting $|x'| < r$, $|x_n| < r$ extends as a graph over $|x'| < 2r$. Continue its local graph along a straight segment in the base. The length at most $3r$ and slope less than $1/3$ keep the height within $2r$. A stopping point would therefore lie inside the larger cylinder, where closedness of the free boundary and $|N_n| \geq 31/32$ allow the implicit function theorem to continue the graph. The sign of N_n stays fixed, and uniqueness for that sign on each vertical line makes these continuations agree. Every free-boundary point in the smaller cylinder is on one of these graphs.

The graph through p has upward inward normal and will be denoted g_+. If the other orientation occurs, its graph g_- lies below g_+. Both graph heights lie within $2r$ over $|x'| < r$, so the vertical segment joining them stays inside the larger cylinder, where each orientation occurs at most once on that line. The open segment starts on the zero side just above g_- and contains no free-boundary point. Thus the gap is in the zero phase, including its intersection with the smaller cylinder. In that smaller cylinder, the inward vertical rays give $v_j > 0$ above g_+ and below g_-. If there is just the graph through p, its slope and $g_+(0) = 0$ keep its height within $r/3$ over $|x'| < r$. Each vertical segment below it inside the smaller cylinder is in the zero phase, since it starts on the zero side and meets no other free boundary there. Figure 5 shows the two-sheet geometry.

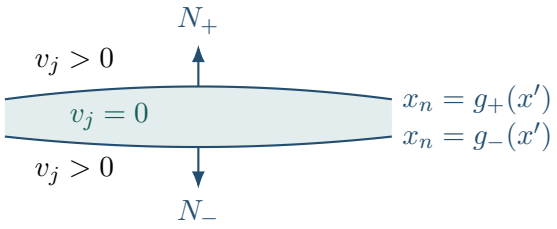


FIGURE 5. A cross-section through two ordered free-boundary sheets. The shaded gap is in the zero phase. The inward normals point into the positive regions on either side.

For derivative bounds, take a point q on one of the graphs over $|x'| < r$, and fix $0 < \rho < r/8$. At any point $x \in B_\rho(q)$, the graph height at x' differs from q_n by at most $\rho/3$. If x is on the inward side, its vertical distance to the graph is at most $4\rho/3 < \ell$, so the ray estimate puts it in the positive phase. Keep v_j on that side in $B_\rho(q)$ and set it to zero on the other side. This is a classical solution in $B_\rho(q)$ with the given sheet as free boundary. In unit-ball coordinates $z = (x - q)/\rho$, divide the solution values by ρ. Its k-th derivatives are then ρ^{k-1} times the corresponding derivatives in the original coordinates. In particular the rescaled Hessian is bounded by ρM. Applying [4, Lemma 3.9] and returning to $B_{\rho/2}(q)$ gives

$$|D^k v_j| \leq C_{n,k} \rho^{2-k} M \max\{1, (\rho M)^{k-2}\}, \qquad k \geq 2.$$

These bounds apply on the retained positive side of the sheet and extend to its one-sided traces by continuity.

To obtain graph derivatives, differentiate $v_j(x', g_j(x')) = 0$ in the coordinates of the cylinder. Writing $v_{j,n} = \partial_n v_j$ and evaluating the derivatives of v_j on the graph, we have

$$\partial_i g_j = -\frac{\partial_i v_j}{v_{j,n}}, \qquad \partial_{il} g_j = -\frac{D^2 v_j(e_i + (\partial_i g_j)e_n,\, e_l + (\partial_l g_j)e_n)}{v_{j,n}} \quad (i, l < n).$$

The denominator has absolute value at least $31/32$. For each higher derivative the chain rule again gives $v_{j,n}$ times that derivative of g_j, plus terms involving only lower derivatives of g_j and derivatives of v_j. This gives uniform bounds at all orders over $|x'| < r$ by induction.

Fix $p_\infty \in \Gamma_v$ and choose $p_j \in \Gamma_{v_j}$ tending to it. Center the cylinders at p_j. Take a subsequence on which the coordinate rotations O_j converge to O and the set of graph orientations in the smaller cylinder is fixed. The upper graph always passes through the chart origin. In these coordinates $\widehat{v}_j(x) = v_j(p_j + O_j x)$. Parametrize the positive side of a graph g_j by

$$\Psi_j(x', s) = (x', g_j(x') + \epsilon s), \qquad |x'| < r/2, \quad 0 \le s \le \rho/4,$$

where $\epsilon = 1$ for the upper graph and -1 for the lower. These collars lie on the indicated positive sides for $s > 0$ by the vertical-ray estimate. Each collar point is within $\rho/2$ of the graph point with the same base coordinate. Thus the derivative bounds apply in this fixed parameter domain. The functions themselves are locally bounded by uniform convergence, and their gradients on a collar are bounded by $1 + Ms$. The bounds on the graphs and on $W_j = \widehat{v}_j \circ \Psi_j$ give smooth convergence on smaller closed collars after a further subsequence, by Arzelà–Ascoli and diagonal extraction over the derivative orders. If Ψ denotes the limiting collar map, the locally uniform convergence identifies the limit of $W_j(x', s)$ with $v(p_\infty + O\Psi(x', s))$.

For small $s > 0$, the bound $\widehat{v}_j(\Psi_j(x', s)) \ge (31/32)s - Ms^2/2$ keeps the limiting collars positive. Local uniform convergence gives zero on the open gap between the limiting graphs, since every point strictly between them is eventually in the prelimit gap. The identities $W_j(x', 0) = 0$ give the zero trace on each limiting graph, including points where the sheets coincide. If only the upper graph is present, local uniform convergence gives zero on its zero side. In a sufficiently small cylinder about the chart origin these descriptions cover the positive phase. Indeed, for $|x'|, |x_n| < a$ with $a < \rho/16$, the upper graph height is bounded by $a/3$. A positive point above it has vertical distance at most $4a/3$ from it. A positive point below the lower graph has the same bound because $g_- \le g_+$. The graph and its positive side therefore give a smooth local solution at every point of a limiting sheet where it is separated from the other. At a contact point the two sides are smooth separately. In both cases the positive-side trace of the gradient is the inward unit normal, by the smooth collar convergence and $D\widehat{v}_j = N_j$ on each graph.

The same convergence controls the physical Hessians up to each sheet. The chain rule gives

$$D^2 W_j = (D\Psi_j)^{\mathsf{T}} (D^2 \widehat{v}_j \circ \Psi_j) D\Psi_j + (\partial_n \widehat{v}_j \circ \Psi_j) \begin{pmatrix} D^2 g_j & 0 \\ 0 & 0 \end{pmatrix},$$
$$\partial_n \widehat{v}_j \circ \Psi_j = \epsilon\, \partial_s W_j.$$

Since $D\Psi_j$ is invertible with determinant ϵ, the C^2 convergence of g_j, W_j gives convergence of $D^2\widehat{v}_j$ at convergent collar coordinates, including $s = 0$ by one-sided traces. Convergence of the rotations gives the same conclusion in the original coordinates. These extractions may be made at a single boundary point at a time, since the locally uniform limit v is already fixed.

If the sheets are separated at every free-boundary point, v is classical everywhere. Its inner stability from [9, Lemma 4.6] therefore makes it a global classical stable solution in the sense of [9, Definition 3.2]. Suppose instead the two limiting graphs touch. At the contact the function $g_+ - g_- \ge 0$ has a zero minimum. Hence $Dg_+ = Dg_-$ and $D^2(g_+ - g_-)$ is positive semidefinite there. The two inward normals are opposite because the tangent planes agree and the positive sides face in opposite directions. With $F(z) = z/\sqrt{1 + |z|^2}$, the inward mean curvatures are $H_+ = -\operatorname{div} F(Dg_+)$ and $H_- = \operatorname{div} F(Dg_-)$. To compute them, extend the upward unit normal

$(-Dg,1)/\sqrt{1+|Dg|^2}$ independently of x_n. Since it has unit length, its derivative in its own direction has zero normal component, so its ambient divergence equals its tangential divergence on the graph. The lower sheet uses the downward normal. Thus at contact

$$H_+ + H_- = -DF(Dg_+) : D^2(g_+ - g_-) \le 0.$$

The sign follows because $DF(z)$ is positive definite:

$$\xi^{\mathsf{T}} DF(z)\xi = \frac{(1+|z|^2)|\xi|^2 - (z\cdot\xi)^2}{(1+|z|^2)^{3/2}} \ge \frac{|\xi|^2}{(1+|z|^2)^{3/2}}.$$

On each smooth positive side, $\sigma = 1-|Dv|^2 \ge 0$ since v is 1-Lipschitz, and $\sigma = 0$ at its boundary. Harmonicity and the boundary Hessian identity of Appendix A give, in any dimension,

$$\partial_{N_\pm}\sigma = 2H_\pm, \qquad \Delta\sigma = -2|D^2 v|^2 \le 0.$$

The zero boundary trace and nonnegativity of σ give $\partial_{N_\pm}\sigma \ge 0$ at the respective sheets. At contact, $H_+ + H_- \le 0$ therefore forces both inward derivatives to vanish. Each smooth sheet has a small tangent ball on its own positive side. Indeed, in tangent-plane coordinates the sheet has quadratic height bound $C|x'|^2$, while the lower cap of a ball of radius s centered at height s has height at least $|x'|^2/(2s)$. Choose s small enough that the ball lies in the positive collar. If σ were not identically zero in this ball, the strong minimum principle and the Hopf lemma would give a strictly positive inward derivative at the contact point. Thus $\sigma = 0$ in each ball, and the Laplacian identity gives $D^2 v = 0$ there.

Write x_* for the contact point. As x approaches x_* within either tangent ball, $v(x) \to 0$ and $Dv(x) \to N_\pm(x_*)$ by the smooth one-sided traces. Since v is affine in each ball, its formula there is $N_\pm(x_*)\cdot(x-x_*)$. Harmonic unique continuation extends each formula to its entire positive component. The components are distinct because their gradients are opposite. By the relative open-and-closed argument of Section 5, each component fills the half-space where its affine formula is positive. These half-spaces are complementary across their common boundary plane through x_*. Hence $v(x) = |N_+(x_*)\cdot(x-x_*)|$. Since $0 \in \Gamma_v$, the common plane passes through the origin.

Proof of Corollary 1.4. In dimension one $u'' = 0$ by harmonicity, and the curvature bound is vacuous. In higher dimensions the Bernoulli rescaling $u_R(y) = R^{-1}u(x_0 + Ry)$ preserves stability by (A.1) and multiplies the Hessian and free-boundary curvatures by R. It suffices to take $R = 1$, $x_0 = 0$. If the Hessian bound failed, choose classical stable u_j in B_2, with $0 \in \Gamma_{u_j}$, such that $\sup_{\Omega_j \cap B_1} |D^2 u_j| \to \infty$, where $\Omega_j = \{u_j > 0\}$. On $\overline{B_{5/4}} \cap \overline{\Omega_j}$, maximize $(5/4 - |x|)|D^2 u_j(x)|$, using one-sided boundary traces. Let p_j be a maximizing point, $d_j = |D^2 u_j(p_j)|$, and $\rho_j = 5/4 - |p_j|$. The weighted Hessian is continuous up to the free boundary from the positive phase, so the maximum exists. It is positive for large j, hence $\rho_j > 0$. The weight is at least $1/4$ on B_1, so the maximum of the weighted Hessian tends to infinity. On $B_{\rho_j/2}(p_j)$ the weight is at least $\rho_j/2$, so maximality also bounds the Hessian there. In formulas,

$$m_j = d_j\rho_j \longrightarrow \infty, \qquad |D^2 u_j| \le 2d_j \quad \text{in } B_{\rho_j/2}(p_j) \cap \Omega_j.$$

We keep a free-boundary point at bounded distance from the Hessian normalization after rescaling. If p_j is in the positive phase, set $s_j = \min\{\operatorname{dist}(p_j, \Gamma_{u_j}), \rho_j/4\}$. The ball $B_{s_j}(p_j)$ is positive and lies in $B_{3/2}$, since $|p_j| + \rho_j/4 \le 5/4$. Applying the interior gradient estimate to the harmonic components of Du_j there gives $s_j d_j \le C_n L_n$ by (6.1). Since $d_j\rho_j \to \infty$, for large j we must have $s_j < \rho_j/4$. The distance to Γ_{u_j} is then attained at a point z_j in the compact ball $\overline{B_{\rho_j/4}(p_j)} \subset B_2$. Hence $d_j|p_j - z_j| \le A_n$ for a dimensional A_n. If $p_j \in \Gamma_{u_j}$, take $z_j = p_j$, for which the same bound holds.

Define $v_j(y) = d_j u_j(z_j + y/d_j)$ and $q_j = d_j(p_j - z_j)$. The scaling preserves stability and gives $D^2 v_j(y) = d_j^{-1} D^2 u_j(z_j + y/d_j)$. In these coordinates the ball $B_{\rho_j/2}(p_j)$ becomes $B_{m_j/2}(q_j)$. Since

$|q_j| \le A_n$, it contains $B_{m_j/2-A_n}$. Thus $0 \in \Gamma_{v_j}$, $|D^2 v_j| \le 2$ on the positive phase in $B_{m_j/2-A_n}$, and $|D^2 v_j(q_j)| = 1$, with the one-sided trace if needed. These are the hypotheses of the compactness argument above. Pass to a subsequence with $q_j \to q$. The local Hausdorff convergence of the closed positive phases and $q_j \in \overline{\{v_j > 0\}}$ give $q \in \overline{\{v > 0\}}$. If $v(q) > 0$, a fixed closed ball about q lies in $\{v > 0\}$. Interior harmonic estimates give uniform Hessian convergence on a smaller ball, so $D^2 v_j(q_j) \to D^2 v(q)$.

If $q \in \Gamma_v$, take the moving charts above centered at points of Γ_{v_j} tending to q. For $a < \rho/16$ fixed, the coordinates of q_j eventually satisfy $|x'|, |x_n| < a$. The upper graph has height at most $a/3$ in absolute value there, and any lower graph lies below it. As in the limit chart, a point on a closed positive side has vertical distance at most $4a/3 < \rho/4$ from its sheet. Thus q_j lies in one of the closed collars. Pass to a subsequence with a fixed choice of side and write the collar coordinates of q_j as (x'_j, s_j) on its graph g_j, with $s_j \ge 0$. The base coordinates tend to zero, and the ray estimate gives $s_j/32 \le v_j(q_j) \to v(q) = 0$. Thus $s_j \to 0$. The vertical chart coordinate is $g_j(x'_j) + \epsilon s_j$ and tends to zero, so the chosen limiting sheet passes through q. The C^2 collar convergence and the Hessian chain rule above therefore give convergence of $D^2 v_j(q_j)$ to the one-sided Hessian trace of v on that sheet at q.

In either case the limit matrix has norm one, since each $|D^2 v_j(q_j)| = 1$. In the classical case, Theorem 1.1 or Corollary 1.3 gives zero Hessian in the positive phase and hence in its boundary trace. In the touching case, the absolute-value profile is affine on both positive sides, so both one-sided Hessian traces are zero as well. This contradicts the norm-one limit.

By Appendix A, the tangential block of the positive-side Hessian is II_Γ with the inward sign convention. Hence $|\mathrm{II}_\Gamma| \le |D^2 u|$ on the trace. The interior bound passes to this trace by continuity and gives the second inequality in (1.3). □

6.2. **Finite-index solutions.**

Proof of Corollary 1.5. Our one-sided smoothness and Bernoulli conditions give classicality in the sense of [8, (1.1)]. Their second-variation form [8, (1.2)] has the same curvature coefficient as (1.2). With the outward normal $\nu = -N$, it is

$$H = -\operatorname{div}_\Gamma \nu = \operatorname{div}_\Gamma N = -D^2 u(N, N).$$

See [4, Appendix A, (A.1)] and the trace identity in Appendix A. Their tests are smooth on the closed positive side with compact support in the ambient domain. Appendix A extends these tests to smooth compactly supported ambient tests. Restriction and extension preserve both terms of Q, since the gradient in the positive phase and the free-boundary trace agree.

Restriction is injective on a negative subspace of ambient tests, since a test with zero restriction also has zero trace and $Q = 0$. Conversely, extend a basis of any finite-dimensional negative subspace of positive-side tests. Restricting the extensions recovers that basis, so their span has the same dimension and is again negative definite. The two indices therefore agree, giving the finite Morse index required in [8, Theorem 1.2]. The disjoint-support argument in Section 1 gives the exterior stability used in their proof [8, Lemma 3.4].

The dimension condition in [8, Theorem 1.2 and (1.3)–(1.5)] is $3 \le n < \min\{n^1_*, n^2_*\}$. These critical dimensions refer to two classes of auxiliary stable solutions. In either class, restricting compactly supported ambient tests to the closed positive side preserves Q, so the scalar stability in [8] implies (1.2). Appendix A then gives inner stability in the same ambient domain.

The first, n^1_*, is the least dimension admitting a non-one-dimensional entire classical stable solution. Theorem 1.1 and Corollary 1.3 give one-dimensionality in dimensions one through six, so $n^1_* \ge 7$.

The second, n^2_*, is the least dimension admitting a non-one-dimensional, one-homogeneous solution that is classical and stable in $\mathbb{R}^n \setminus \{0\}$. A candidate v extends continuously by $v(0) = 0$,

since $v(r\theta) = rv(\theta)$ and v is bounded on the unit sphere. Points with $v > 0$ approach the origin along a ray, so 0 lies on the free boundary of the extension. The cone classification in Section 2 uses inner deformations supported away from the origin and gives a half-space solution in dimensions two through six. Every solution in dimension one depends on a single coordinate. Thus $n_*^2 \geq 7$, and

$$3 \leq n \leq 6 < 7 \leq \min\{n_*^1, n_*^2\}.$$

For a solution depending on one linear coordinate, rotations about that coordinate axis fix u. For any other solution, [8, Theorem 1.2] gives the two alternatives, the monotonicity, and the zero set in the statement. The exterior-ball solution is invariant under rotations about every line through its center.

In the graph alternative, g is differentiable on $(1, \infty)$ with $g'(r) > 0$ there. In particular $g(r) > 0$ for $r > 1$, by the mean value theorem and $g \geq 0$. For $|\omega| = 1$, the upper free boundary at $(r\omega, g(r))$ contains the radial tangent $(\omega, g'(r))$ and sphere tangents $(\tau, 0)$ for $\tau \perp \omega$. Their horizontal projections span $\mathbb{R}^{n-1}$, so the horizontal projection restricted to the smooth free boundary has invertible derivative. The inverse function theorem makes the upper height smooth near $r\omega$, hence g is smooth for $r > 1$. Monotonicity gives a right limit at 1. If this limit were positive, choose $s > 0$ below it. The points $(r\omega, s)$ would be in the zero set for $r > 1$ close to 1. By continuity so would (ω, s), contrary to the zero-set formula and $g(1) = 0$. Thus g is continuous at 1. The finite limit at infinity from [8, Theorem 1.2] is at least $g(2) > 0$. □

7. Hessian energy and neck exclusion

In dimension six, the integral of $|D^2u|^6$ is invariant under Bernoulli scaling. Marked necks carry a fixed positive amount of this energy. We bound it on a smaller ball by combining height closeness to a two-sided profile on a larger ball with a Hessian bound on the smaller ball.

Proposition 7.1 (Hessian energy at small height excess)**.** *Let u be classical stable in $B_{8R}(z_0) \subset \mathbb{R}^6$, with $|Du| \leq 1$ in the positive phase there. For a unit vector e, set $V(x) = |e \cdot (x - z_0)|$ and*

$$\varepsilon = \left(R^{-8} \int_{B_{8R}(z_0)} (u - V)^2 \right)^{1/2}.$$

For sufficiently small ε, every ball $B_{2r}(z) \subseteq B_{2R}(z_0)$ satisfies

$$\int_{\Omega \cap B_r(z)} |D^2u|^2 \leq C\varepsilon^{1/4} r^4. \tag{7.1}$$

If also $\sup_{\Omega \cap B_r(z)} |D^2u| \leq K/r$, then

$$\int_{\Omega \cap B_r(z)} |D^2u|^6 \leq CK^4 \varepsilon^{1/4}. \tag{7.2}$$

The smallness threshold and C are dimensional.

The distributional Laplacian records the area of the free boundary. For a smooth test χ compactly supported in the ambient domain, harmonicity gives $\operatorname{div}(\chi Du) = Du \cdot D\chi$ in the positive phase. The only boundary flux is on Γ, where $\chi \partial_\nu u = -\chi$ for the outward normal ν. Thus

$$\int_\Gamma \chi = -\int_\Omega Du \cdot D\chi. \tag{7.3}$$

By the weak-gradient formula in Section 1, the right-hand side equals $\langle \Delta u, \chi \rangle$. Thus, in dimension six, $\Delta u = \mathcal{H}^5 \lfloor \Gamma$.

Proof of Proposition 7.1. Set $x = z_0 + Ry$ and $\widehat{u}(y) = R^{-1}u(x)$. The rescaled comparison profile is $|e \cdot y|$. Since $dx = R^6\,dy$ and $u(x) - V(x) = R(\widehat{u}(y) - |e \cdot y|)$,

$$\varepsilon^2 = \int_{B_8} (\widehat{u}(y) - |e \cdot y|)^2\,dy.$$

The gradient and stability bounds are preserved by (A.1). Also $B_{2r}(z)$ becomes $B_{2r/R}((z - z_0)/R) \subseteq B_2$. Work first in the normalized coordinates, using u, V, z, r for the normalized data. Write $v = u - V$ and $w = |D^2u|$ in the positive phase for the Hessian norm.

By the weak-gradient formula in Section 1 and the hypothesis $|Du| \le 1$ in the positive phase, the ambient weak gradient of u has norm at most one almost everywhere. This gives the 1-Lipschitz bound on the convex ball B_8. Mollify on a neighborhood of any segment compactly contained in B_8, integrate the gradient bound along the segment, and pass to the locally uniform limit. As V is 1-Lipschitz, v is 2-Lipschitz. Fix $x \in B_6$ with $m = |v(x)| > 0$, and take $s = \min\{1, m/4\}$. For $y \in B_s(x)$, $|v(y)| \ge m - 2s \ge m/2$. Since $B_s(x) \subset B_8$,

$$\varepsilon^2 \ge \int_{B_s(x)} |v|^2 \ge \tfrac{1}{4}|B_1|\,m^2 s^6.$$

For small ε, $m \ge 4$ is impossible because it would give $s = 1$ and $\varepsilon^2 \ge 4|B_1|$. Using $s = m/4$ now gives $m^8 \le C\varepsilon^2$. Thus

$$\|v\|_{L^\infty(B_6)} \le C\varepsilon^{1/4}.$$

To control the gradient error, first bound the measure $\mu = \Delta v$ on B_5. For a nonnegative cutoff $\vartheta \in C_c^\infty(B_6)$ equal to one on B_5, (7.3) gives

$$\Delta u(B_5) \le \int_\Gamma \vartheta = -\int_\Omega Du \cdot D\vartheta \le \int |D\vartheta|.$$

For $x = y + te$ with $y \in e^\perp$, the weak gradient of V is $\operatorname{sgn}(t)e$ almost everywhere. For a smooth compactly supported test ϕ, integrating $-\operatorname{sgn}(t)\partial_t\phi(y + te)$ over the two half-lines gives $2\phi(y)$. Integration in y yields $\Delta V = 2\mathcal{H}^5\lfloor e^\perp$. The second measure has mass $2\mathcal{H}^5(e^\perp \cap B_5)$ on B_5. Since both measures are nonnegative, their sum bounds $|\mu|$ there and gives $|\mu|(B_5) \le C$.

Choose $0 \le \chi \le 1$ smooth, equal to one on B_4, and compactly supported in B_5. We test $\mu = \Delta v$ with $\chi^2 v$, as in the gradient estimate in the proof of [9, Proposition 4.12]. The test is Lipschitz with compact support in B_5. Mollifying it gives smooth tests supported in B_5 that converge uniformly and in H^1. These two convergences pass to the integrals against μ and Dv, respectively. Expanding the gradient of the test and using $D(v^2) = 2vDv$ then gives

$$\begin{aligned}\int \chi^2|Dv|^2 &= -\int \chi^2 v\,d\mu - \tfrac{1}{2}\int D(\chi^2)\cdot D(v^2)\\ &= -\int \chi^2 v\,d\mu + \tfrac{1}{2}\int v^2\Delta(\chi^2).\end{aligned}$$

The absolute value of the measure term is at most $\|v\|_{L^\infty(B_6)}|\mu|(B_5) \le C\varepsilon^{1/4}$, and the other term is at most $C\|v\|^2_{L^2(B_8)} = C\varepsilon^2$. Since $\chi = 1$ on B_4, for $\varepsilon \le 1$ this proves $\int_{B_4}|Dv|^2 \le C\varepsilon^{1/4}$.

On Ω, (5.2) with $q = DV = \pm e$ gives $h_e^2 \le (1 + 4/30)|Dv|^2$ almost everywhere. Thus the defect on B_4 satisfies

$$E_e(B_4)^2 = 4^{-6}\int_{\Omega\cap B_4} h_e^2 \le C\varepsilon^{1/4}.$$

In particular $E_e(B_4) \le C\varepsilon^{1/8}$. Choose the height threshold small enough that this is at most ε_0 in Theorem 1.2. Applying that theorem on B_4 and squaring its bound gives

$$\sup_{\Omega\cap B_2} h_e^2 \le C\varepsilon^{1/4}.$$

For $B_{2r}(z) \subseteq B_2$, take $\eta = 1$ on $B_r(z)$, smooth and compactly supported in $B_{2r}(z)$, with $|D\eta| \le C/r$. Lemma 3.1 gives the Robin inequality for h_δ and $h_\delta \Delta h_\delta \ge w^2/75$. Thus (4.1) with $G = h_\delta$ yields

$$\tfrac{1}{75} \int_{\Omega \cap B_r(z)} w^2 \le \int_\Omega \eta^2 h_\delta \Delta h_\delta \le \int_\Omega (h_e^2 + \delta^2) |D\eta|^2.$$

The term with δ^2 vanishes as $\delta \downarrow 0$. The remaining integral is at most $C\varepsilon^{1/4} r^{-2} |B_{2r}(z)| \le C\varepsilon^{1/4} r^4$, since $|B_{2r}(z)| = |B_1|(2r)^6$.

In the original coordinates, $D^2\widehat{u}(y) = RD^2u(z_0 + Ry)$ and $dy = R^{-6}\,dx$. With $\widehat{z} = (z - z_0)/R$, this gives

$$\int_{\{\widehat{u}>0\} \cap B_{r/R}(\widehat{z})} |D^2 \widehat{u}|^2 = R^{-4} \int_{\Omega \cap B_r(z)} |D^2 u|^2.$$

The normalized bound has radius factor $(r/R)^4$, so multiplying by R^4 gives (7.1). If $|D^2u| \le K/r$ on the original positive ball, then

$$\int_{\Omega \cap B_r(z)} |D^2 u|^6 \le (K/r)^4 \int_{\Omega \cap B_r(z)} |D^2 u|^2 \le CK^4 \varepsilon^{1/4},$$

which is (7.2). □

For the neck application, consider the normalized contradiction setting of [9, Assumption 5.3] in $n = 6 < n_*$, where the dimension bound is from Section 2. In this setting $|Du| \le 1$ and $|D^2u| \le 1$ in Ω. Let $\mathcal{Z}$ be the marked centers of [9, Definition 5.4], and let $\eta_0 > 0$ be the dimensional threshold in [9, (5.2)]. For a marked center z, set

$$I_z(s) = \int_{\Omega \cap B_s(z)} |D^2 u|^6, \qquad r_*(z) = \inf\{s > 0 : I_z(s) \ge \eta_0^6\}.$$

The bound $I_z(s) \le |B_1| s^6$ gives $r_*(z) > 0$. Marked centers are selected only from the finite-radius classes in [9, Definition 5.4], so $r_*(z) < \infty$. The integral $I_z(s)$ is continuous in $s > 0$: the ball indicators converge away from the limiting sphere, which has Lebesgue measure zero, and the integrands are bounded by one on a fixed larger ball. It is less than η_0^6 below $r_*(z)$, and radii with $I_z(s) \ge \eta_0^6$ approach $r_*(z)$. Hence $I_z(r_*(z)) = \eta_0^6$. With $M = 1$, the local Hessian bound in [9, Lemma 5.7] gives $|D^2u| \le K/r_*(z)$ on $\Omega \cap B_{r_*(z)}(z)$ for a dimensional $K > 0$.

For a direction e at scale $8R$, write $M^e_{8R,z_0} = (8R)^{-8} \int_{B_{8R}(z_0)} (u - V)^2$ for the height term in [9, Definition 7.6, (7.8)]. The square of the combined excess $\widetilde{\boldsymbol{E}}_{z_0}(8R; e)$ is the sum of this term and a supremum along the ridge of the nonnegative Gaussian Weiss deficit from [9, Lemma 7.4]. Consequently

$$\varepsilon = 8^4 \sqrt{M^e_{8R,z_0}} \le 8^4 \widetilde{\boldsymbol{E}}_{z_0}(8R; e).$$

For a marked z with $B_{2r_*(z)}(z) \subseteq B_{2R}(z_0)$, the Hessian bound and (7.2) from Proposition 7.1 give

$$\eta_0^6 = I_z(r_*(z)) \le CK^4 \varepsilon^{1/4}.$$

Let $\overline{\varepsilon}$ be the height threshold in Proposition 7.1. If the minimized combined excess at $8R$ is less than

$$8^{-4} \min\left\{ \overline{\varepsilon}, \left(\frac{\eta_0^6}{CK^4} \right)^4 \right\},$$

take a direction with combined excess below this bound. Then $\varepsilon < \overline{\varepsilon}$ and $CK^4\varepsilon^{1/4} < \eta_0^6$, excluding every marked ball satisfying the containment. In particular the containment is automatic for $z = z_0 \in \mathcal{Z}$ with $r_*(z_0) \le R$.

Appendix A. Scalar and inner stability

Let u be classical in an open set A, with the normal and curvature conventions of Section 1. We will use smooth functions and vector fields on the closed positive side with compact support in A. To extend one of these functions or vector fields, cover the boundary part of its support by finitely many boundary charts and localize it by a smooth partition of unity. Flatten each boundary piece and use smooth extension from the closed half-space, componentwise for a vector field. Multiplying by cutoffs that are one on the localized supports keeps the extensions in their charts. Adding the interior pieces gives an ambient extension with compact support in A.

On Γ, differentiating $Du = N$ along any tangent vector τ gives $D^2u\,\tau = D_\tau N$. Since $|N| = 1$, $D_\tau N$ is tangent, so the mixed entries of the Hessian vanish by symmetry. For an orthonormal tangent basis (τ_i), the trace identity is

$$0 = \Delta u = D^2u(N,N) + \sum_i \langle D_{\tau_i} N, \tau_i \rangle = D^2u(N,N) + H.$$

Consequently $D^2u\,N = -HN$.

Let Φ_t be a smooth family of diffeomorphisms with $\Phi_0 = \mathrm{id}$, equal to the identity outside a fixed compact subset of $D \Subset A$. Set $u_t = u \circ \Phi_t^{-1}$, $\Omega_t = \Phi_t(\Omega)$, $\Gamma_t = \Phi_t(\Gamma)$, and $J(t) = J(u_t; D)$. The velocity at the moving spatial point is $X_t = \dot\Phi_t \circ \Phi_t^{-1}$, and the outward normal speed of Γ_t is $V_t = X_t \cdot \nu_t$. Differentiating $u_t \circ \Phi_t = u$ gives $\dot u_t = -Du_t \cdot X_t$ in Ω_t, where the dot on u_t denotes differentiation at a fixed spatial point. On Γ_t we take its positive-side trace. There $Du_t = -|Du_t|\nu_t$, since u_t vanishes on Γ_t and is positive on the inward side. Thus $\dot u_t = |Du_t|V_t$ on Γ_t. At $t = 0$ we write $X = X_0$, $\psi = \dot u_0 = -Du \cdot X$, so $V_0 = \psi$ on Γ.

Since Φ_t fixes the complement of D, it maps D onto itself. Pull the energy integral over $\Omega_t \cap D$ back to $\Omega \cap D$. The time derivatives of the integrands are supported on the fixed compact set where the deformation acts. In the pulled-back coordinates these derivatives are bounded there for small t, by smoothness of Φ_t and of u up to the positive-side boundary. Thus differentiation under the integral is valid. Differentiating $\dot\Phi_t = X_t \circ \Phi_t$ in space gives $D\dot\Phi_t = (DX_t \circ \Phi_t)D\Phi_t$. The determinant derivative therefore gives $\dot j_t = (\operatorname{div} X_t) \circ \Phi_t\, j_t$ for $j_t = \det D\Phi_t$. The chain rule in the pulled-back integrand and the Jacobian derivative give, in moving coordinates,

$$J'(t) = \int_{\Omega_t \cap D} \big(2Du_t \cdot D\dot u_t + \operatorname{div}\big((1 + |Du_t|^2)X_t\big)\big).$$

The flux fields $(1 + |Du_t|^2)X_t$ and $\dot u_t Du_t$ vanish near ∂D, so only the free boundary contributes. The divergence theorem and Green's identity give

$$\begin{aligned} J'(t) &= 2\int_{\Omega_t \cap D} Du_t \cdot D\dot u_t + \int_{\Gamma_t \cap D} (1 + |Du_t|^2)V_t \\ &= -2\int_{\Omega_t \cap D} \Delta u_t\, \dot u_t + \int_{\Gamma_t \cap D} (1 - |Du_t|^2)V_t. \end{aligned}$$

Indeed, the boundary term from the Dirichlet integral is $2\dot u_t \partial_{\nu_t} u_t = -2|Du_t|^2 V_t$. At $t = 0$ the harmonic and Bernoulli conditions give $J'(0) = 0$.

To differentiate $J'(t)$ at zero, pull the two integrals back by Φ_t to the fixed volume and surface domains. The same compact support and smoothness justify differentiation on both domains. At $t = 0$ the factors Δu in the bulk and $1 - |Du|^2$ on the boundary vanish. Hence derivatives of the velocities and of the volume and surface Jacobians drop out. In the bulk the derivative of $\Delta u_t \circ \Phi_t$ at zero is $\Delta\psi + X \cdot D(\Delta u) = \Delta\psi$. The material derivative of Du_t at zero is $D\psi + D^2u\,X$ by the chain rule in the interior. The formula extends to its positive-side trace on Γ. Hence the boundary coefficient has derivative

$$-2N \cdot (D\psi + D^2u\,X) = 2(\partial_\nu \psi - H\psi).$$

Here $-N = \nu$ and $N \cdot D^2 u\, X = (D^2 u\, N) \cdot X = -HN \cdot X = H\psi$. Using $V_0 = \psi$ and integrating by parts once more yields

$$J''(0) = -2 \int_\Omega \psi\, \Delta\psi + 2 \int_\Gamma \psi\, \partial_\nu \psi - 2 \int_\Gamma H\psi^2 = 2Q(\psi).$$

The integrands have compact support in D. Since ψ extends to a smooth compactly supported ambient test, (1.2) implies inner stability.

Conversely, suppose u is inner stable and fix a signed $\phi \in C_c^\infty(A)$. Where $Du \neq 0$ in the positive phase, the choice $X = -\phi Du/|Du|^2$ gives $-Du \cdot X = \phi$. To use this construction on the entire support, we remove the interior critical set by cutoffs. Only finitely many components of A meet $\operatorname{supp}\phi$, since finitely many balls contained in A cover this compact support, and each ball lies in a single component. We work on one such component at a time and add the resulting inequalities. If the component contains no free boundary, its contribution to Q is the nonnegative Dirichlet term. On the remaining components, u is nonconstant on each positive component. Indeed, a relative boundary point of a positive component has value zero, as in Section 5. A component on which u is a positive constant therefore has no such boundary point and fills its ambient component. In dimension one the scalar inequality holds directly because the tangential trace H is zero, so we take $n \geq 2$ below.

The critical set $\mathcal{C} = \{Du = 0\} \cap \Omega \cap \operatorname{supp}\phi$ on the components under consideration is compactly contained in Ω. Indeed, a limit on Γ would contradict the continuous positive-side gradient trace with $|Du| = 1$. Stratify $\mathcal{C}$ by the least order $m \geq 2$ for which a derivative of u is nonzero at the point. Such an order is finite by real analyticity of nonconstant harmonic functions. At a point of order m choose an order-$(m-2)$ derivative f of u for which $D^2 f \neq 0$. This Hessian is symmetric and trace free by harmonicity. If it had rank one, its single nonzero eigenvalue would give a nonzero trace. Thus two components of Df have linearly independent gradients at the point. These components are derivatives of u of order $m-1$, so they vanish at every point in the same stratum. The implicit function theorem puts the part of the stratum in this neighborhood in a smooth graph of codimension two. Take a countable chart cover in each stratum. Exhaust the whole graph in each chart by compact subsets of the graph inside the chart, so these compact pieces cover $\mathcal{C}$.

Each compact chart piece admits a smooth cutoff equal to one in a neighborhood of the piece and with arbitrarily small H^1 norm, supported inside Ω. In flattened coordinates, multiply a tangential cutoff by a radial cutoff in the two normal variables. For normal radius r, let it be one for $r \leq \epsilon^2$, zero for $r \geq \epsilon$, and a smooth function of $\log(\epsilon/r)/|\log\epsilon|$ in between. Its derivative is bounded by $C/(r|\log\epsilon|)$. The squared L^2 norm and the tangential derivative cost are $O(\epsilon^2)$. The normal derivative cost is bounded by

$$\frac{C}{|\log\epsilon|^2} \int_{\epsilon^2}^{\epsilon} \frac{dr}{r} = \frac{C}{|\log\epsilon|}.$$

Changes of coordinates on the compact support preserve these bounds up to fixed constants. For each j, choose the cutoff for chart piece i with H^1 norm at most $2^{-i}/j$. The neighborhoods where the cutoffs equal one cover $\mathcal{C}$, so finitely many suffice. Sum that finite collection and compose with a smooth function that is zero at zero, equals one on $[1,\infty)$, and has bounded derivative and values in $[0,1]$. This gives $\eta_j \in C_c^\infty(\Omega)$ equal to one near $\mathcal{C}$, with $\|\eta_j\|_{H^1} \leq C \sum_{i \geq 1} 2^{-i}/j = C/j$ by the triangle inequality and the chain rule.

Set $\phi_j = \phi(1 - \eta_j)$. Its trace on Γ is ϕ, since η_j has compact support in Ω. The product rule gives $\|\phi_j - \phi\|_{H^1} = \|\phi\eta_j\|_{H^1} \leq C_\phi \|\eta_j\|_{H^1} \to 0$. The field $X_j = -\phi_j Du/|Du|^2$ is smooth where $Du \neq 0$. It extends by zero across critical points in $\operatorname{supp}\phi$ because $\eta_j = 1$ near $\mathcal{C}$, and across critical points outside $\operatorname{supp}\phi$ because ϕ vanishes in a neighborhood of each such point. Extend X_j smoothly and with compact support across Γ as above. Its flow realizes $\psi = \phi_j$ in the

second-variation identity, so $Q(\phi_j) \geq 0$. The bulk term converges by the H^1 convergence and the boundary term is unchanged along this sequence. Thus $Q(\phi) \geq 0$ for all signed tests.

For later use, let $y = x_0 + Rx$ with $R > 0$, and set $u_R(x) = R^{-1}u(y)$. Then $D_x u_R(x) = D_y u(y)$ and $D_x^2 u_R(x) = RD_y^2 u(y)$, so the Bernoulli conditions and any gradient bound are preserved. For a test ϕ in the rescaled domain, put $\Phi(y) = \phi((y - x_0)/R)$. Then $D_x\phi = RD_y\Phi$ and $dx = R^{-n}dy$. The inward normal is $N_R(x) = N(y)$. For a tangent vector X at corresponding points, $D_X N_R(x) = RD_X N(y)$ by the chain rule. The principal curvatures are the eigenvalues of this tangential operator, so they and their sum H are multiplied by R. As $dS_x = R^{-(n-1)}dS_y$, the bulk and boundary terms in Q acquire the same factor

$$Q_{u_R}(\phi) = R^{2-n}Q_u(\Phi). \tag{A.1}$$

Thus scalar stability, and hence inner stability, is preserved.

Appendix B. Weighted local curvature estimate

The local Hessian bound also yields a weighted estimate for third derivatives. A regularized power of the Hessian norm controls these derivatives through its Laplacian, and the Bernoulli condition bounds its normal derivative on the free boundary.

Theorem B.1 (Local curvature growth)**.** *Let u be classical stable in $B_{4R}(x_0) \subset \mathbb{R}^6$, with $x_0 \in \Gamma$, and write $w = |D^2u|$. Then*

$$\int_{\Omega\cap B_R(x_0)} (w^{42/11} + w^{-2/11}|D^3u|^2) + \int_{\Gamma\cap B_R(x_0)} w^{31/11} \leq CR^{24/11}. \tag{B.1}$$

The constant C is dimensional. The weighted integrand is defined to be zero where $w = 0$, and the boundary integral uses the one-sided trace.

Proof of Theorem B.1. Write $B_s = B_s(x_0)$ throughout the proof. Corollary 1.4 applied with radius $2R$ gives $w \leq C/R$ on $\Omega \cap B_{2R}$. The same bound holds on $\Gamma \cap B_{2R}$ by continuity from the positive side. The rescaled function $v(y) = (2R)^{-1}u(x_0 + 2Ry)$ is classical in the ball of radius two centered at zero, with $0 \in \Gamma_v$ and $Dv(y) = Du(x_0 + 2Ry)$. Thus (6.1) gives $|Du| \leq C$ in $\Omega \cap B_{3R}$. Choose a smooth cutoff $0 \leq \chi \leq 1$ equal to one on B_{2R} and compactly supported in B_{3R}, with $|D\chi| \leq C/R$. By (7.3),

$$\mathcal{H}^5(\Gamma \cap B_{2R}) \leq \int_\Gamma \chi = -\int_\Omega Du \cdot D\chi \leq CR^{-1}|B_{3R}| \leq CR^5.$$

Consequently

$$\int_{\Omega\cap B_R} w^{42/11} \leq (C/R)^{42/11}|B_R| \leq CR^{6-42/11} = CR^{24/11},$$
$$\int_{\Gamma\cap B_R} w^{31/11} \leq (C/R)^{31/11}\mathcal{H}^5(\Gamma \cap B_{2R}) \leq CR^{5-31/11} = CR^{24/11}.$$

For the remaining term, set $T = D^2u$, $\beta = 10/11$, $W_\delta = w^2 + \delta^2$, and $f_\delta = W_\delta^\beta$, where $0 < \delta \leq R^{-1}$. The components $T_{jk} = \partial_j\partial_k u$ satisfy $\Delta T_{jk} = \partial_j\partial_k\Delta u = 0$. Differentiating $W_\delta = \sum_{j,k} T_{jk}^2 + \delta^2$ gives

$$\partial_i W_\delta = 2\sum_{j,k} T_{jk}\partial_i T_{jk}, \qquad \Delta W_\delta = 2\sum_{i,j,k}(\partial_i T_{jk})^2 = 2|DT|^2.$$

By Cauchy–Schwarz, $|DW_\delta|^2 \le 4w^2|DT|^2$. The gradient term in the chain rule has coefficient $\beta(\beta-1)<0$, so this estimate gives

$$\begin{aligned}\Delta f_\delta &= \beta W_\delta^{\beta-1}\Delta W_\delta + \beta(\beta-1)W_\delta^{\beta-2}|DW_\delta|^2\\ &\ge 2\beta W_\delta^{\beta-2}(\delta^2+(2\beta-1)w^2)|DT|^2\\ &\ge 2\beta(2\beta-1)W_\delta^{\beta-1}|DT|^2.\end{aligned}$$

The last inequality follows from $0<2\beta-1<1$.

To estimate the boundary flux, let S be the tangential block of T. By Appendix A, $D_XN=SX$ for tangent X, the mixed entries vanish, and $T\nu=-H\nu=HN$. The normal diagonal entry is therefore $-H$, so $\operatorname{tr}T=0$ gives $\operatorname{tr}S=H$ and the orthogonal blocks give $w^2=|S|^2+H^2$. Differentiating $T\nu=HN$ along a tangent vector X and pairing with a tangent vector Y gives

$$D^3u(X,\nu,Y)+\langle TD_X\nu,Y\rangle=(XH)\langle N,Y\rangle+H\langle D_XN,Y\rangle.$$

Using $D_X\nu=-SX$ and $\langle N,Y\rangle=0$, we obtain $D^3u(\nu,X,Y)=\langle(S^2+HS)X,Y\rangle$. For an orthonormal tangent basis $(\tau_i)_{i=1}^5$, the positive-side trace of $\partial_\nu\Delta u=0$ gives

$$D^3u(\nu,\nu,\nu)=-\sum_{i=1}^5 D^3u(\nu,\tau_i,\tau_i)=-|S|^2-H\operatorname{tr}S=-w^2.$$

At the point of evaluation choose a fixed Euclidean orthonormal frame $(\tau_1,\dots,\tau_5,\nu)$. In this frame T has blocks S and $-H$, so differentiating the squared norm gives

$$\tfrac12\partial_\nu(w^2)=\operatorname{tr}\big(S(S^2+HS)\big)+(-H)(-w^2)=\operatorname{tr}(S^3)+H|S|^2+Hw^2.$$

If λ_i are the eigenvalues of S, then $|\operatorname{tr}(S^3)|\le\sum_i|\lambda_i|^3\le|S|^3$. As $|S|,|H|\le w$, this gives $|\partial_\nu(w^2)|\le 6w^3$. For $w>0$, $W_\delta\ge w^2$ and $\beta-1<0$, so

$$|\partial_\nu f_\delta|\le 6\beta W_\delta^{\beta-1}w^3\le 6\beta w^{2\beta+1}.$$

At $w=0$, the bound $|\partial_\nu(w^2)|\le 6w^3$ gives $\partial_\nu f_\delta=0$.

Choose a smooth cutoff $0\le\kappa\le1$ equal to one on B_R and compactly supported in B_{2R}. Scaling a fixed cutoff gives $|D\kappa|\le C/R$ and $|\Delta\kappa|\le C/R^2$. The divergence of $\kappa Df_\delta-f_\delta D\kappa$ is $\kappa\Delta f_\delta-f_\delta\Delta\kappa$. The field is supported away from ∂B_{4R}, so the divergence theorem gives

$$\int_\Omega\kappa\Delta f_\delta=\int_\Omega f_\delta\Delta\kappa+\int_\Gamma\kappa\partial_\nu f_\delta-\int_\Gamma f_\delta\partial_\nu\kappa.$$

On the support of the cutoff, $w\le C/R$ and $\delta\le R^{-1}$ give $f_\delta\le CR^{-2\beta}$ and $|\partial_\nu f_\delta|\le CR^{-(2\beta+1)}$ on Γ. The absolute values of the three terms on the right are bounded by $CR^{6-2\beta-2}$, $CR^{5-(2\beta+1)}$, and $CR^{5-2\beta-1}$, respectively, using the volume and area bounds for B_{2R}. All three exponents are $4-2\beta=24/11$. Since $\kappa\ge0$ and $\kappa=1$ on B_R, the lower bound for Δf_δ yields

$$2\beta(2\beta-1)\int_{\Omega\cap B_R}W_\delta^{\beta-1}|DT|^2\le\int_\Omega\kappa\Delta f_\delta\le CR^{24/11}.$$

On $\Omega\cap B_R\cap\{w=0\}$, the regularized weight equals $\delta^{-2/11}$. Thus the integral of $|DT|^2$ over this set is at most $CR^{24/11}\delta^{2/11}$ for every such δ, and $DT=0$ there almost everywhere. On $\{w>0\}$, the weight increases to $w^{-2/11}$ as $\delta\downarrow0$. Monotone convergence now gives

$$\int_{\Omega\cap B_R}w^{-2/11}|D^3u|^2\le CR^{24/11}$$

with the stated zero-set convention. Adding the two bounds for the powers of w proves (B.1). □

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